

國立陽明交通大學
資訊管理研究所
碩士論文
初稿

Department of Information Management
National Yang Ming Chiao Tung University
Master Thesis

經濟逆境或社群影響？通膨下永續消費的驅動因素分析

Economic Headwinds or Social Tides? Analyzing Drivers of
Sustainable Consumption Engagement Online During Inflation

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中華民國 一一四年六月
June 2025

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A Thesis Draft
Submitted to Department of Information Management
College of Management
National Yang Ming Chiao Tung University
in partial Fulfilment of the Requirements
for the Degree of
Master of Science
in
Information Management

June 2025

Taiwan, Republic of China

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摘要

近年永續意識雖逐步提升，但消費者實際選購永續商品時，仍普遍面臨價格考量與理念實踐的衝突。此矛盾在通貨膨脹加劇時期更為顯著，也凸顯一個關鍵問題：究竟哪些因素能真正驅動人們在日常消費中落實永續行動？

本研究提出了一個整合性分析框架，結合經濟心理學與社會影響理論，並納入有限理性觀點，探討經濟壓力如何影響永續參與度。透過分析 Reddit 平台上的公開討論數據，系統性解構用戶生成內容中的四大行為訊號：情感傾向、議題關注度、社群互動模式與經濟壓力感知，同時對照客觀經濟指標。研究發現，相較於傳統經濟數據，個體的影響力與社群互動模式更能有效預測對永續議題的參與程度。本研究在理論上有助釐清個人決策與社會影響的交互作用機制，實務層面則能輔助經濟壓力情境下的永續策略，以提升永續倡議的實際影響力。

關鍵字: 永續參與、經濟心理學、社會影響、有限理性、使用與滿足理論、可解釋式人工智慧

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Abstract

In recent years, sustainability awareness has grown significantly. Governments and individuals alike have begun to place greater emphasis on sustainable development. However, when it comes to actual purchasing behavior, many consumers still face a dilemma between supporting eco-friendly products and managing the rising cost of living. During inflationary periods, this trade-off becomes even more pronounced, raising questions about what factors drive or hinder engagement with sustainability content.

This study proposes a multi-module analytical framework grounded in multiple behavioral theories, including economic psychology, social influence, bounded rationality, and uses and gratifications theory. Drawing on large-scale Reddit data and macroeconomic indicators, the system extracts key behavioral signals such as sentiment, topic relevance, peer influence, and contextual economic stress. These features are then used to model engagement patterns. Results reveal that individual influence and social cues have greater predictive power than economic indicators alone. The framework provides both theoretical insights into the interplay between individual constraints and social dynamics, and practical implications for effective sustainability communication in times of economic uncertainty.

Keyword: Sustainable consumption; Inflation; Social media engagement; Social influence; Economic headwinds; Sentiment analysis; Reddit; Explainable AI

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Chapter 1

Introduction

1.1 Background

The shift toward sustainable development has become one of the major global goals in recent decades. A key part of this transition involves changing consumer behavior, encouraging people to choose products and services that are more environmentally friendly and better for the planet in the long term [1]. In recent years, public awareness of sustainability issues has increased due to greater access to information and more open discussions. However, this doesn't always lead to action. There's still a clear gap between what people say they care about and what they actually do when making purchases [2]. Even when consumers express strong environmental values, those values often don't fully translate into sustainable choices in daily life.

One major reason for this gap is the current macroeconomic challenges, especially the issue of inflation. Rising prices affect basic living costs and often force consumers to prioritize affordability over other product attributes [3, 4]. Studies show that people tend to cut back on non-essential spending during inflation and actively search for cheaper alternatives [5, 6]. This creates strong economic headwinds that directly challenge sustainable consumption. Compared to conventional products, sustainable alternatives are usually more expensive or at least perceived that way [7, 8]. For many people, that price difference becomes a practical barrier. As a result, understanding how consumers navigate the trade-offs between their sustainability goals and these pressing economic constraints within an inflationary environment becomes critically important.

At the same time, social media platforms (like Reddit, which is the focus of this study) have become powerful spaces where people shape opinions, discuss values, and influence each other's decisions [9, 10]. Social media can be a positive force for advancing sustainability.

It helps spread information, builds community support, and encourages people to share their sustainable choices [11, 12]. Through peer influence and online conversations, it can foster a collective **social tide** that supports pro-environmental attitudes and encourages more people to take action [13, 14].

However, the role of social media isn't always straightforward. These platforms also serve as channels for amplifying anxieties, including those related to economic hardship driven by inflation [15]. In online discussions, it's not uncommon to see complaints about the cost of eco-friendly products, which can reinforce the idea that sustainable choices are financially unrealistic. In this way, social media can act as both a support and a barrier, at once encouraging people to be more environmentally aware and making them more aware of the challenges that come with that. This creates a **dynamic tension** between two forces: the desire to live more sustainably and the economic reality of inflation. How these forces interact in online environments—and how they shape people's willingness to engage with sustainability is the main issue this research tries to explore.

While past research has explored how either social interaction or economic pressure shapes consumer behavior, most studies focus on a single perspective, leaving key interdependencies underexplored.

In the domain of **social influence**, studies have shown that peer interaction and social media platforms can shape sustainability attitudes, spark behavior change, and amplify community narratives [16, 17]. However, these works often examine engagement in stable conditions and overlook how social signaling operates under economic stress. Additionally, prior analyses typically focus only on post-level cues such as likes or comments, while *author-level influence*—how credible contributors drive participation—remains under-investigated [18].

On the other hand, from the economic perspective, researchers have examined how inflation, affordability, and value-action gaps influence sustainable consumption [19, 20]. These studies reveal that rising prices can suppress green behavior, but often rely on structured survey data and neglect how people spontaneously express cost-related stress or coping strategies in real-world discourse [21, 22].

Despite progress in both areas, few studies have integrated these perspectives to examine how **economic pressure and social influence jointly shape sustainability discourse**, especially in large-scale, unstructured environments like Reddit. This research bridges that gap by combining both theoretical strands into a unified framework. Through layered modeling and explainable AI, we analyze how emotional tone, peer dynamics, and macroeconomic context interact to drive online engagement. In doing so, this study moves beyond isolated predictors and provides a richer, behaviorally grounded understanding of sustainability participation under inflation.

1.2 Motivation and Research Problems

This research is mainly motivated by a need to better understand the complicated and sometimes conflicting factors that shape how people respond to sustainability during times of economic stress. Inflation can make it even harder for consumers to act in line with their values. Even if someone believes in sustainability, they may feel forced to prioritize basic needs or cheaper options when their budget is tight [2, 19]. This situation shows a conflict between two sides: on one hand, there are **rational economic concerns** like affordability and cost; on the other hand, there are **social and emotional influences** such as personal values, social identity, or what people around them are doing and saying [5, 3, 16].

Instead of just pointing out that this conflict exists, we want to go further by looking into how these factors actually show up in real-world online behavior. While prior work has used surveys or controlled experiments to measure sustainability attitudes, few studies analyze spontaneous social media expressions that reveal how users justify, debate, or dismiss sustainable choices under financial pressure [22, 21]. The idea is to explore which kinds of posts, messages, or signals are most effective at encouraging people to stay engaged with sustainability even when financial pressure is high. A key part of this is finding out whether the “**social tide**” (including peer support, shared values, and online community dynamics [11, 17]) can help balance or even overcome the negative effects of “**economic headwinds**”. More importantly, the study aims

to identify exactly **which aspects of online interactions** are most helpful in doing that.

To examine these ideas in a concrete way, this study focuses on Reddit—a social media platform that is known for active discussions and a high degree of user anonymity. On Reddit, users can upvote or downvote posts, creating a simple but meaningful indicator of engagement [18]. In this study, **online engagement** is mainly measured by a post’s score (net upvotes), which reflects how much a post resonates with the community. Based on this, the concepts of **economic headwinds** and **social tides** are turned into observable features drawn from each post. These features fall into four main categories: Content Features, Economic Signals, Sentiment & Emotion, and Social & Interaction Signals.

The goal is not primarily prediction accuracy, but to utilize predictive modeling as an **analytical tool** to assess the **relative importance** of different feature categories in explaining variations in online engagement, thereby identifying the **key drivers**.

Based on the research motivations above, this thesis proposes two research questions to structure the subsequent analysis:

1. *How do economic pressures and social influences interact to shape sustainability engagement online during inflation?* This research starts from the idea that consumers often face a tension between what they care about and what they can realistically afford—especially during times of inflation. On one side, rising costs create economic headwinds, making people more price-sensitive and less willing to choose sustainable options that are often seen as more expensive [5, 6]. On the other side, social media platforms like Reddit can generate strong social tides: communities that promote sustainability, share eco-friendly habits, and create peer pressure to engage in pro-environmental behavior [11, 12]. However, these platforms can also amplify worries about affordability. Threads discussing sustainability often include complaints about high prices or doubts about whether sustainable choices are really worth it. This question looks at the combined effect of these opposing forces. Specifically, it asks whether social influence can help buffer the negative impact of economic concerns, or whether economic pressure ends up dominating how people engage with sustainability topics online. The focus is on how this interaction

shows up in actual conversations and engagement metrics, like post scores.

2. *What are the key drivers of online engagement with sustainability topics during inflation?*

After understanding the broader push and pull between economic and social factors, the next step is to look more closely at what actually drives engagement in online discussions. This question aims to identify which types of information or features play the biggest role in getting people to interact with sustainability-related content on Reddit. We group potential drivers into four categories: structural and content-related features, emotional tone and sentiment, social cues, and economic signals. By using predictive modeling and explainable AI tools like SHAP, this study looks at which categories, and which specific features within them are the strongest drivers of engagement during times of economic stress. The goal is to figure out what actually helps people connect with sustainability content when it's not easy to care.

1.3 Research Objectives and Contributions

Understanding how people engage with sustainability topics online, especially during challenging economic periods is important for both academic research and real-world application. This study aims to develop and apply a multi-layered analytical framework to identify what drives online engagement with sustainability during inflation. The approach combines different types of signals: content features, emotional cues (including nuanced intent and sentiment using LLMs), economic references, and social interaction patterns. Through layered modeling and explainable AI (XAI) methods like SHAP, the study explores how much each factor matters and how they work together.

The main contribution is theoretical: bringing together economic psychology and social influence theory to better explain consumer behavior in a high-pressure, digital environment. Past studies often looked at economic or social factors on their own. This research connects the two, offering a more complete view. From a methods perspective, it demonstrates how real-world online data and advanced modeling techniques can help uncover the patterns behind value-driven

behavior. On the practical side, the results are expected to provide useful insights for marketers, policymakers, and sustainability advocates—especially those looking to engage people through messages or platforms that don’t rely purely on price incentives. Social motivation may offer an alternative path forward when budgets are tight.

1.4 Thesis Outline

The remainder of this thesis is structured as follows. Chapter 2 will review the relevant literature pertaining to our study. Chapter 3 will discuss the system framework of our proposed mechanism. The experiments conducted in our research will be showcased in Chapter 4. Chapter 5 will present the analytical results and key observations from the modeling. Finally, Chapter 6 will conclude the thesis by summarizing the key findings and contributions, acknowledging limitations, and suggesting directions for future research.

Chapter 2

Related Literature

This chapter reviews key theories and past research to build the foundation for this study. The goal is to better understand the different forces that shape how people engage with sustainability, especially during periods of inflation and within the fast-moving world of social media. This chapter connects perspectives from economics, psychology, and communication to form a coherent framework that supports later analysis.

We first look at how people's choices are shaped by rational cost considerations and psychological reactions to economic stress, particularly in inflationary times. Concurrently, we focus on how online communities and social influence shape perceptions, attitudes, and behaviors related to sustainability. Together, these two areas help explain the tension many people face: they may value sustainability, but still feel constrained by price or social pressure. Understanding this tension, and how social dynamics might help reduce it, is key to this study. The insights from this chapter will guide how variables are defined and measured in Chapter 3, and how the results are interpreted later on.

2.1 Economic Psychology

Economic psychology offers a useful perspective for understanding how inflation shapes consumer thinking and decision-making. When prices rise, people tend to focus more on basic needs and affordable options. In this process, long-term values like sustainability often take a back seat [23, 24]. These shifting priorities may show up in online discussions through increased mentions of cost, savings, and concerns about making ends meet.

One common reaction to inflation is emotional. Many people experience what's known as inflation aversion—a negative feeling toward rising prices, sometimes accompanied by blaming governments or companies [25]. This frustration can erode trust and change how people shop.

On social media, it might appear in the form of complaints, sarcasm, or negative sentiment toward expensive products, including those that are eco-friendly.

To understand these behaviors more deeply, Dual-Process Theory helps explain how people make decisions under pressure [26, 27]. This theory describes two systems of thinking. System 1 is fast, intuitive, and emotionally driven—especially activated when people are stressed or overwhelmed. In contrast, System 2 is more deliberate and analytical, weighing longer-term considerations like environmental impact. In inflationary contexts, people may rely more on System 1, choosing the cheapest option to feel immediate relief, while the slower, value-based reasoning of System 2 is often sidelined. This internal tension between saving money now and supporting sustainability goals, is likely to show up in how people write about their choices online.

Inflation also tends to activate certain cognitive biases that pull people further away from sustainable behavior. For example, money illusion can cause people to focus on nominal prices instead of real purchasing power, leading to distorted judgments about what products are “worth it” [28]. Loss aversion makes price increases feel more painful than equal savings feel rewarding, which can discourage spending on higher-priced sustainable items [29]. Present bias is another factor—it pushes people to care more about immediate savings than future environmental benefits [30]. These kinds of mental shortcuts often surface in online comments, where users justify avoiding sustainable products or express frustration over their cost.

It’s also important to remember that not everyone responds to inflation in the same way. Demographic factors, cultural background, and overall consumer confidence can affect how people react—even in similar economic conditions [31, 25]. This diversity in reactions can be observed in the range of opinions and tones seen in Reddit posts and comments. Some users may express strong anti-sustainability sentiment, while others try to find ways to stay green without overspending.

Overall, economic psychology highlights several key ideas like inflation aversion, dual-process thinking, and common decision-making biases—that help explain why people often avoid sustainable choices when economic pressure builds. This study draws on these concepts

to interpret large-scale online discussions. These ideas help guide our analysis of sentiment and price-related discussions, offering insight into how people balance affordability and sustainability under economic stress.

2.2 Bounded Rationality and Budget Constraints

While economic psychology highlights how inflation affects people emotionally, everyday decision-making is also shaped by cognitive limits and financial realities. Two important ideas: bounded rationality and budget constraints, help explain how people make trade-offs when money is tight.

Bounded rationality, introduced by Herbert Simon, suggests that people rarely make perfectly rational decisions. Instead, they rely on simplified thinking, aiming for options that are “good enough” rather than ideal [32]. Under inflation, this becomes even more pronounced. Price fluctuations, shrinkflation, and unclear value comparisons make it harder to process all available options. As a result, consumers are more likely to rely on heuristics—mental shortcuts that help reduce effort but can also lead to suboptimal outcomes [33].

At the same time, budget constraints grow tighter as inflation erodes purchasing power [6]. People often reprioritize, focusing on essential needs first. Even if they care about sustainability, they may choose lower-cost alternatives simply because that’s what their budget allows [34]. In this context, sustainability may be seen as valuable but non-essential—something to postpone for better times.

Together, these two pressures push consumers toward simpler decision rules. For example, some might develop the mindset that “sustainable = expensive,” making green products easier to dismiss without much evaluation. Others may default to familiar choices, avoiding the cognitive load involved in comparing eco-labels or understanding environmental claims. This can reinforce status quo bias, where people stick to what they know to avoid added effort [35].

These tendencies can also be observed in real-world online discussions. On platforms like Reddit, for example, some users express frustration with endless price comparisons, avoid trying

new “green” brands, or explain their choices by saying they simply can’t afford sustainable options right now.

2.3 Social Influence Theory (SIT)

Social Influence Theory (SIT) helps explain how people’s decisions and attitudes are shaped by others—whether it’s through direct interaction or simply observing what others do [36, 37]. The theory highlights two main forms of influence: normative influence, which comes from the desire to fit in and gain social approval, and informational influence, which happens when people treat others’ opinions as useful guidance, particularly in uncertain situations. These forces are relevant in social media spaces, where both types of influence are constantly in play.

In today’s online environments, social influence shows up in many forms. People often rely on eWOM (electronic word-of-mouth), online reviews, and the general tone of community interactions when making decisions [38]. For example, informational influence can come from detailed product reviews, personal experiences, or advice threads, while normative influence shows up in things like upvotes, high comment counts, or alignment with community values [39]. On platforms like Reddit, for instance, the upvote/downvote system directly signals normative approval or disapproval, while the threaded discussion format facilitates extensive informational exchange and learning [40].” Past research has also found that factors like how information is presented, emotional intensity, or perceived social connection can all affect how credible or useful a post seems [41, 42, 43].

During times of uncertainty—like periods of inflation—these social cues may matter even more. When people feel unsure about their finances, they’re more likely to look at what others are doing or recommending before making a choice [44]. In the case of sustainability, social influence can help people decide whether it’s worth spending a bit more on environmentally friendly products. Several studies show that even when money is tight, peer behavior and perceived social norms can strongly shape whether people stick with sustainable values or shift away from them [45, 46]. In this way, a strong social tide within online communities might help

counterbalance the pressure to prioritize price alone.

This study uses Social Influence Theory to better understand how social dynamics affect sustainability discussions in an inflationary context. It provides a foundation for analyzing how peer behaviors, shared norms, and collective sentiments shape what gets noticed, discussed, or embraced in online sustainability conversations.

2.4 Uses and Gratifications Theory (U&G)

Uses and Gratifications Theory (U&G) offers a helpful perspective for understanding why people actively choose to engage with certain types of media. Rather than treating audiences as passive receivers, U&G emphasizes that individuals are purposeful users who interact with media to fulfill specific needs, such as gaining information, connecting socially, expressing identity, or managing emotions [47, 48]. This user-centered approach is especially relevant to platforms like Reddit, where users are not just reading content, they are actively posting, responding, and building communities around shared interests [49, 50].

In the context of sustainability-related discussions, particularly during inflationary periods, U&G helps explain why users choose to participate despite financial constraints. Even when budget pressures increase, many individuals continue to seek out content that addresses their evolving needs. Some users may be looking for practical tips on how to live sustainably without overspending, such as advice on affordable green alternatives or strategies for reducing environmental impact in cost-effective ways [51]. Others turn to these platforms for social connection and emotional validation, hoping to share their experiences, find encouragement from like-minded individuals, or gain support for maintaining sustainable habits amid economic stress [52]. Online spaces like Reddit also allow people to express frustrations about rising costs, voice anxieties about environmental or financial concerns, or celebrate small victories in sustainable living. These expressions serve both personal and communal purposes, helping individuals process emotion while reinforcing their identity as environmentally conscious users even when constrained by real-world limitations.

When these psychological or social needs are met, users may feel more motivated to stay involved in discussions. For instance, someone who finds a useful post and receives supportive feedback may become more likely to revisit the community, share more ideas, and maintain a positive attitude toward sustainability. On the other hand, if a post fails to resonate or lacks relevance to users' current needs, it may receive less engagement, regardless of how informative or well-intentioned it is.

In this study, U&G provides a lens for understanding the intent behind different types of posts and comments in sustainability-related discussions during inflation. By analyzing how users seek advice, share experiences, express frustration, or promote personal values, we can begin to identify which needs are most actively driving engagement. This approach helps explain why certain types of messages attract more attention than others—because they speak to the specific gratifications that users are seeking in an economically stressful environment.

2.5 Summary

This chapter reviewed four theoretical perspectives that together form the conceptual basis for this study. From the economic domain, Economic Psychology and Bounded Rationality explain how consumers under inflationary pressure may rely on emotional shortcuts and simplified heuristics, often deprioritizing sustainability when cost becomes a constraint. From the social domain, Social Influence Theory (SIT) and Uses and Gratifications Theory (U&G) provide insight into how peer dynamics, normative cues, and individual motivations shape online participation and message reception.

Together, these theories highlight the dual influence of **rational limitations and social dynamics** in shaping sustainability engagement online. They guide the selection of features and design of analysis modules in the following chapters, offering a framework for understanding how these forces manifest in real-world social media discourse during inflationary periods.

Chapter 3

The System Framework

3.1 Overview

This chapter presents the theoretical and analytical framework that guides this study's investigation into how financial pressure and social dynamics jointly influence sustainability engagement on social media. As shown in Figure 3.1, the system architecture is built on a progression from data acquisition to feature extraction, factor interpretation, and final modeling, each stage informed by behavioral theory.

At the foundation, this study integrates two types of data: user-generated discourse from Reddit and macroeconomic indicators reflecting inflationary trends. Through keyword filtering, a process guided by the concept of **Bounded Rationality**—the system identifies thematically relevant content that reflects how users simplify decisions under cognitive and financial constraints.

Subsequent information extraction modules are aligned with specific theoretical lenses. The **Customer Preference Module**, informed by **Uses and Gratifications Theory**, captures affective tone and topic orientation to reflect users' motivational needs during stress. The **Social Interaction Module**, grounded in **Social Influence Theory**, analyzes post structure and author reputation to evaluate normative and informational peer effects. The **Macroeconomic Alignment Module** leverages **Economic Psychology** to examine how emotional or engagement patterns correspond with broader economic indicators.

These extracted signals are then synthesized in two stages: **Factor Investigation**, which identifies key behavioral drivers of engagement, and **Engagement Modeling**, which tests their predictive value. This dual structure supports both theoretical interpretation and practical classification, enabling the system to explain—and anticipate—how online sustainability discourse

evolves under economic stress.

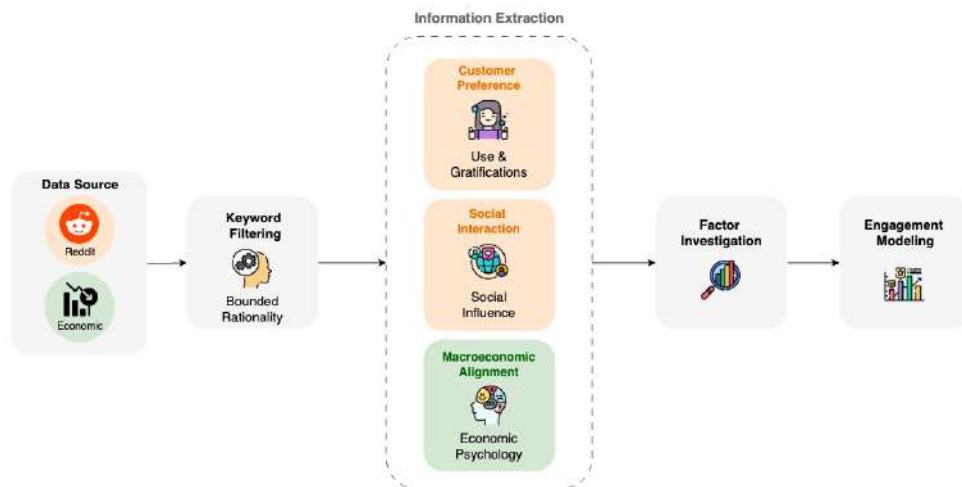


Figure 3.1: Theoretical framework and analytical flow of the proposed system.

Building on this theoretical foundation, we developed a five-module system (Figure 3.2) to transform Reddit discourse and macroeconomic signals into actionable insights. The pipeline begins with the Data Construction Module, followed by three core analytic modules: Customer Preference (sentiment and topic analysis), Social Interaction (post and creator signals), and Macroeconomic Alignment (external trend analyzing). These feed into the final Engagement Driver Identification Module.

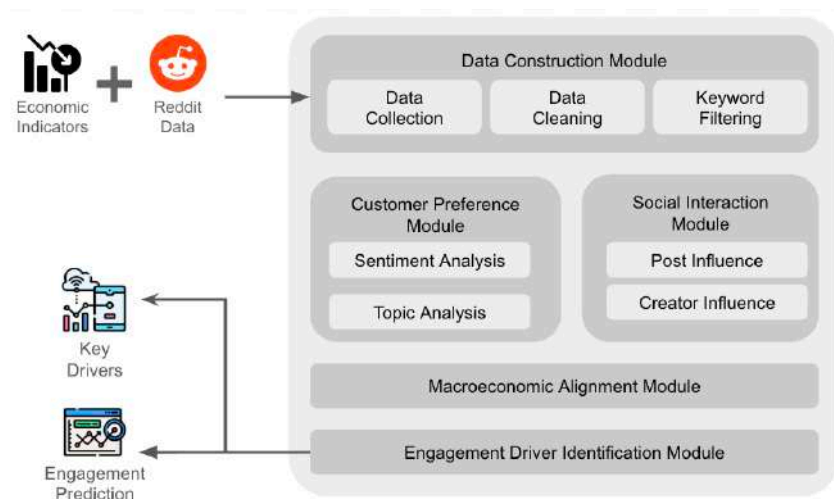


Figure 3.2: System architecture for analyzing online engagement during inflation.

This system produces dual outputs. On the one hand, it supports **engagement prediction**, enabling classification of high-impact content and providing practical value for communication

strategy. On the other hand, it identifies **key behavioral drivers**, such as author reputation, emotional tone, and affordability concerns, which are grounded in established theories. These theoretical insights enrich our understanding of how users respond to sustainability messages during periods of economic stress.

Rather than treating context as an external variable, each module operationalizes theory, and informed constructs directly within the social media corpus. This structure reflects a normative methodological contribution: embedding behavioral logic into feature extraction, modeling, and interpretation to uncover both predictive signals and explanatory mechanisms. The system thus offers a layered and theory-grounded approach to understanding public participation in sustainability discourse during turbulent economic times.

3.2 Data Construction

To examine how inflation and social influence affect sustainability engagement, this module constructs a research-ready dataset from Reddit discussions and macroeconomic indicators. It integrates Reddit discourse and macroeconomic indicators to form a research-ready corpus that balances breadth (coverage of related themes) and depth (semantic and behavioral cues). Reddit's topic-based structure and support for long-form discourse make it well-suited for capturing authentic public opinions. By including both issue-oriented (e.g., r/sustainability) and behavior-oriented (e.g., r/Frugal) subreddits, the dataset ensures topical and demographic diversity, enhancing its representativeness.

The module draws on the principle of **bounded rationality**, which suggests that individuals rely on heuristics under cognitive limitations. Here, we use keyword filtering functions as a heuristic mechanism, guiding the system to identify thematically relevant content without processing the entire discourse space. Classifying posts into conceptual themes also supports later stage modular analysis and interpretability.

Data construction involves three core steps: data collection, data cleaning, and keyword filtering. Each step ensures that the dataset is not only clean and consistent, but also analytically

aligned with the research objectives and theoretical framework.

3.2.1 Data Collection

This module integrates Reddit discussions and macroeconomic indicators to construct a research-ready dataset. Reddit’s topic-based structure, public accessibility, and support for long-form discussions make it well-suited for observing value-driven behavior. Reddit data is sought to capture authentic, user-generated expressions reflecting both the ”social tides” of community influence and individual perceptions of ”economic headwinds.” The data collection is designed to cover discussions from domains central to the research questions: sustainability, consumer behavior, and economic influence.

Meanwhile, macroeconomic indicators are selected to provide an objective, quantifiable context for the online discussions. The rationale for including these indicators (such as CPI, PCEPI, ICS, and unemployment rates from official U.S. sources) is to enable subsequent analysis of the relationship between prevailing economic conditions and the sentiments or topics emerging from online discourse. This allows for an examination of how public online sentiment aligns with, or diverges from, broader inflationary trends and economic realities.

3.2.2 Data Cleaning

To ensure the dataset was suitable for natural language processing and modular analysis, standard cleaning procedures were applied:

- Removal of non-informative content (e.g., [deleted], [removed], bot messages)
- Filtering by minimum text length to exclude noise and memes
- Text normalization: lowercase conversion, punctuation removal, and lemmatization

‘’ These procedures standardize linguistic inputs for sentiment and topic modeling while preserving semantic fidelity. Non-English posts and duplicated records were removed to improve consistency and model performance.

3.2.3 Keyword Filtering

To isolate relevant discussions for analysis, a multi-theme keyword filtering strategy was implemented. This step enables the modular design of subsequent analytical components by tagging each post according to its thematic alignment.

Keywords were grouped into the following three domains:

- **Sustainable Product & Market:** terms such as "eco-friendly," "biodegradable," "green brand," etc.
- **Economic Context:** terms related to inflation, costs, affordability, and economic pressure
- **Consumer Behavior:** terms referencing purchase decisions, spending habits, budgeting, etc.

Seed keywords were first manually selected based on literature and subreddit exploration. To enhance robustness and colloquial coverage, Large Language Models (specifically GPT-4o) were employed to expand each category into a diverse list of terms and expressions. Final keyword sets were applied to the cleaned Reddit dataset using regex-based matching.

Posts were retained if they matched any of the above themes, and each record was tagged accordingly. These thematic labels facilitate targeted analysis in the following modules (e.g., economic framing vs. sustainability framing) and support stratified evaluations of sentiment, influence, and engagement.

3.3 Customer Preference Analysis

To understand how consumers feel about sustainability and how these feelings are influenced by inflation, the **Customer Preference Module** is designed. This module focuses on analyzing user-generated content to explore people's attitudes, emotions, and discussions around sustainable consumption under economic stress. Rather than looking only at the presence of certain keywords, this part of the system pays attention to how users express themselves: the tone of

their messages, the intentions behind their posts, and the recurring themes in what they talk about. These goals are closely related to the ideas introduced in Chapter 2, especially from *Economic Psychology*, which describes emotional reactions to inflation, and *Uses and Gratifications Theory*, which helps explain why people participate in online discussions in the first place.

This module integrates two primary analytical techniques: Sentiment Analysis and Topic Analysis. Together, they allow us to examine both how people feel and what they choose to focus on when discussing sustainability during inflation.

3.3.1 Sentiment Analysis

Sentiment analysis is used to explore the emotional tone of users' posts. It helps us determine whether consumers perceive sustainable products as essential and worthwhile or as non-essential and costly during economic downturns.

Given the large scale of social media data, efficiently extracting emotional insights becomes a practical challenge. While Large Language Models (LLMs) are effective for fine-grained interpretation, applying them to over 200,000 posts is computationally demanding. To balance scale and depth, this study adopts a two-stage sentiment analysis strategy.

In the first stage, a sentiment classification model is used to label posts as positive, neutral, or negative. This provides a general view of emotional trends across key themes, such as sustainability, pricing concerns, and product preferences. The aim is to capture broad patterns in how users respond emotionally to these topics.

To further understand how these emotional tones are received by the community, we introduce a measure called the **average sentiment score** for each theme—sentiment pair. Reddit's score function (i.e., net upvotes) serves as a proxy for how much a post resonates with its audience.

Let T represent the discussion theme (e.g., sustainability, consumer behavior, or economic influence), and S denote the sentiment category (positive, neutral, or negative). For each pair (T, S) , we define:

- $D_{T,S}$: the set of posts assigned to theme T and sentiment S ,
- $N_{T,S}$: the number of posts in $D_{T,S}$,
- score_i : the Reddit score of post i .

Then, the average sentiment score $\mu_{T,S}$ is computed as:

$$\mu_{T,S} = \frac{1}{N_{T,S}} \sum_{i \in D_{T,S}} \text{score}_i$$

This value reflects how well each type of emotional expression aligns with broader community sentiment. A higher $\mu_{T,S}$ may indicate that certain sentiments receive more validation or support, linking back to social mechanisms such as normative approval and informational relevance, as discussed in the U&G and SIT frameworks.

In the second stage, we apply a large language model to a smaller, representative subset of the data. This allows for more detailed labeling of emotions—such as frustration, hope, or anxiety—and underlying intent, including seeking advice, sharing experiences, or expressing dissatisfaction. These insights offer a closer look at how users articulate their responses when navigating the trade-offs between economic concerns and environmental values.

Together, this two-stage approach provides both coverage and depth. It enables us to capture widespread emotional reactions related to inflation and sustainability while also identifying more subtle, intent-driven expressions within the community.

3.3.2 Topic Analysis

While sentiment analysis reveals how users feel, topic analysis helps us understand what they are actually discussing. In this module, we aim to uncover the dominant discussion patterns that emerge in posts related to sustainability, consumption, and economic stress. These patterns can highlight whether users are more focused on product quality, affordability, or systemic environmental issues during inflationary periods.

For this task, we apply Latent Dirichlet Allocation (LDA), a commonly used model for identifying clusters of words that frequently appear together. Each cluster forms a topic, which we can then interpret based on its most representative words and example posts. The idea is to move from isolated terms to meaningful groups of discussion, such as affordability, product quality, eco-label skepticism, or budget-friendly lifestyle tips.

The preprocessing stage ensures that the extracted topics are both meaningful and distinct. This involves steps such as tokenization, removal of noisy or non-informative words, and detection of common multi-word expressions. We also carefully select the number of topics by balancing interpretability with statistical coherence.

Once topics are generated, we interpret them using both salient keywords and representative examples. Human-readable labels are assigned manually, sometimes with the assistance of large language models to help ensure clarity and consistency. These topics provide the foundation for later analyses that connect sentiment, influence, and engagement behavior. They also offer insights into the types of informational or emotional needs users express, aligning with the Uses and Gratifications framework discussed earlier.

3.4 Social Interaction Analysis

Social influence plays a critical role in shaping consumer decision-making, particularly in the context of sustainable consumption. In the digital age, platforms like Reddit have emerged as important arenas where users exchange opinions, react to content, and collectively shape public discourse. Through mechanisms such as upvoting, commenting, and sharing, certain posts and users gain disproportionate visibility, allowing specific narratives to be amplified or challenged.

To understand how such influence operates under the dual pressure of sustainability concerns and economic uncertainty, this module focuses on two complementary dimensions: the influence of individual posts and the aggregated influence of content contributors (authors). These two perspectives enable us to explore not only which content gains traction, but also which voices tend to shape community engagement over time.

3.4.1 Post Influence Analysis

While a large number of Reddit discussions mention sustainability-related issues, only a subset generate high levels of attention or interaction. Understanding what distinguishes high-engagement content from lower-impact posts is essential to identifying the content features and narrative strategies that resonate with users, particularly under inflationary conditions.

This analysis adopts a descriptive approach to examine content-level influence, with Reddit’s **score** (net upvotes) serving as the primary proxy for post visibility and community approval. Given the right-skewed distribution of scores, with most posts receiving modest engagement and a few achieving disproportionately high attention, we define the influence of a post I_{post} as:

$$I_{\text{post}} = \text{score}_{\text{post}} \quad (3.1)$$

Where $\text{score}_{\text{post}}$ is the final score of the Reddit submission.

To account for the heterogeneity in engagement levels, content is stratified into percentile-based *influence tiers*, such as low, medium, and high impact groups. These groupings allow for systematic comparison of content features across different engagement levels.

The comparison involves both surface-level features (e.g., whether the post is a submission or comment, whether it contains external links, text length) and semantic attributes derived from prior modules. These include the thematic category assigned during keyword filtering (e.g., Sustainability, Consumer Behavior, Economic Influence) as well as sentiment attributes from the two-stage sentiment analysis pipeline. In particular, deep sentiment and intent labels derived from a sampled subset via large language models (LLMs) allow for a finer-grained understanding of emotional tone and communicative purpose across different influence levels.

In addition, we consider potential interaction effects between variables. For example, the effect of sentiment on engagement may vary depending on whether the content is a submission or a comment, or whether it is long-form or concise. Identifying such patterns provides insight into how various content attributes combine to influence audience response.

By recognizing which characteristics are consistently associated with higher engagement,

this module contributes to a deeper understanding of how sustainability-related discourse spreads in online communities. Rather than attempting to predict influence through machine learning, this analysis highlights meaningful patterns that inform both theoretical insights and practical strategies for effective sustainability communication. The results from this component serve as a foundation for subsequent exploration of author-level influence.

3.4.2 Author Influence Analysis

In addition to the influence of individual posts, the role of authors in shaping sustainability discussions is also essential. Certain users in Reddit communities consistently contribute content that receives high levels of engagement, helping to set the tone of conversation and amplify particular narratives. Unlike follower-based platforms, Reddit measures influence more implicitly through community upvotes and participation rather than formal connections. This makes it particularly well-suited for studying sustained influence at the author level.

To capture this form of influence, we propose a framework that quantifies user impact based on community reception of their content. Rather than simply counting how often a user posts, the framework emphasizes the quality and engagement level of their contributions. Two metrics are introduced:

1. **Author Influence Score (AIS):** This core metric is designed to capture the *overall accumulated positive engagement* an author receives from the community across all their contributions. It is calculated as a weighted sum of the score (upvotes minus downvotes) received by the author’s submissions (S_a) and comments (C_a). The specific formula is:

$$AIS_{author} = W_{submission} \cdot \sum_{s \in S_a} \text{score}(s) + W_{comment} \cdot \sum_{c \in C_a} \text{score}(c) \quad (3.2)$$

where $\text{score}(s)$ and $\text{score}(c)$ are the scores of individual submissions and comments, respectively. Differential weights ($W_{submission}$ and $W_{comment}$) are applied to potentially reflect the differing nature or visibility of submissions versus comments (The specific weights used in the experiment are detailed in Chapter 4, Section 4.4.2). This score rep-

resents the author’s *total impact and visibility* within the scope of the data.

2. **Average AIS:** To provide a measure of *average content impact density*, mitigating potential bias towards high-volume contributors, the Average AIS is calculated as:

$$AIS_{author}^{avg} = \frac{AIS_{author}}{N_{submissions} + N_{comments}} \quad (3.3)$$

where $N_{submissions}$ and $N_{comments}$ are the total counts of submissions and comments by the author, respectively. This metric reflects the typical influence generated per contribution.

Based on these two metrics, a set of **top influential authors** can be identified. For example, the top 1% of users by AIS may be considered highly influential, while thresholds on average AIS can help identify high-impact users regardless of volume.

This module sets the stage for subsequent analyses, where we examine how these top authors differ from the broader population in terms of content type, sentiment, and sustainability themes. Understanding these differences offers insight into who and how shapes the conversation within economically stressed online communities.

3.5 Macroeconomic Alignment Module

This module investigates the extent to which online sustainability discussions on Reddit align with broader macroeconomic conditions. By examining correlations between internal Reddit metrics and external economic indicators, the goal is to uncover whether user sentiment, discussion volume, and engagement levels reflect—or respond to—real-world economic trends such as inflation and consumer confidence.

Rather than studying Reddit activity in isolation, this module evaluates its responsiveness to macroeconomic stressors. For example, it tests whether spikes in inflation correlate with declines in sustainability-related sentiment or engagement, or whether consumer confidence aligns with more optimistic discourse.

Conceptually, this alignment relies on the integration of **temporally aggregated Reddit-based metrics** and **external macroeconomic indicators**. On the Reddit side, key metrics include monthly discussion volumes by theme (sustainability, consumer behavior, and economic influence), average sentiment scores and positive sentiment ratios within each theme, as well as average engagement scores derived from post-level Reddit scores. These internal signals are then compared with key economic indicators. The primary focus is on inflation, measured by the year-over-year change in the Consumer Price Index (CPI), and consumer confidence, reflected in the Index of Consumer Sentiment (ICS). Additional contextual indicators, such as the national unemployment rate and the Personal Consumption Expenditures Price Index (PCEPI), are incorporated to provide a broader view of the economic environment.

The primary statistical technique used is the Pearson correlation coefficient, which quantifies linear associations between time-aligned variables. To assess directional dependencies, Granger causality tests are applied to selected variable pairs after ensuring stationarity (via ADF test and differencing as needed). These analyses offer insight into whether economic indicators can predict changes in online sentiment or engagement.

The specific implementation procedures and data transformations are detailed in Chapter 4, while results and interpretations are discussed in Chapter 5.

3.6 Engagement Driver Identification Module

The final analytical component of this framework—the **Engagement Driver Identification Module**—aims to integrate insights from prior analyses to systematically determine which factors most strongly influence online engagement with sustainability-related content during inflationary periods. The objective is not simply to build a high-performing predictive model. Instead, predictive modeling is used as an **analytical tool** to assess the **relative importance** of distinct information categories in shaping engagement outcomes, thereby addressing **RQ2**.

The final module of this framework aims to integrate findings from earlier modules, to systematically determine which factors most strongly influence online engagement with sustainability-

related content. The objective is not simply to build a high-performing predictive model. Instead, predictive modeling is used as an **analytical tool** to assess the **relative importance** of distinct information categories in shaping engagement outcomes.

This design is informed by the broader context outlined in Chapter 1, which highlights two opposing forces: economic headwinds that increase price sensitivity and reduce pro-environmental willingness, and social tides that amplify sustainability norms, emotional resonance, and peer influence within online communities. RQ1 focuses on how these two forces interact. In turn, RQ2 seeks to identify the **strongest drivers of engagement** under such conflicting pressures.

To operationalize this investigation, we group potential engagement predictors into four conceptual categories: (1) structural and content-related features, (2) emotional tone and sentiment, (3) social cues (e.g., author reputation, social validation signals), and (4) economic signals (e.g., inflation, consumer sentiment). A layered modeling strategy is then adopted to isolate and evaluate the **marginal contribution of each category**, both independently and in combination.

The model layers are defined as follows:

- **L1_Baseline:** Basic post metadata (e.g., type, text length, subreddit), serving as a foundational benchmark.
- **L2-1_Customer Preference:** Adds sentiment and topic-based features reflecting individual attitudes and values.
- **L2-2_Social Influence:** Includes features derived from post-level and author-level influence, such as average scores, author engagement, and structural signals.
- **L3_Econ (Economic Headwind):** Adds external macroeconomic indicators—including year-over-year inflation (CPI), consumer sentiment index (ICS), and unemployment rate—on top of L1. This layer captures environmental constraints faced by consumers.
- **L4_Customer Preference + Social Influence (Social Tide):** Combines L2-1 and L2-2 to capture the full effect of content quality and social dynamics, representing the normative and emotional pressures found in online platforms. It reflects the "social tide" described in Chapter 1.

- **L5_Full Model:** Integrates all feature categories—baseline, customer preference, social influence, and economic context. This model offers the most comprehensive view of engagement drivers and their interplay.

The predictive task is framed as a **binary classification problem**: determining whether a given Reddit post falls within the top 10% by final score (high engagement). Model performance is evaluated using standard metrics (e.g., F1-score, AUC); however, the primary analytical goal is explanatory. SHAP (SHapley Additive exPlanations) is employed to examine which categories—and which specific variables within them—are most influential in driving engagement during inflation.

Chapter 4

Experiments

This chapter details the experimental procedures used to implement the multi-module analytical framework introduced in Chapter 3. Following the logic of the five system modules, this chapter focuses on the technical workflows, preprocessing routines, feature engineering steps, and modeling strategies applied to Reddit data and macroeconomic indicators. The goal is to offer a transparent view of how theoretical constructs were translated into measurable variables and predictive analyses.

4.1 Data Construction

As outlined in the Data Construction Module in Chapter 3, this stage establishes the foundational dataset for all subsequent analyses. Informed by the theory of bounded rationality and aligned with the study’s three core themes (sustainability, consumer behavior, and economic pressure), this step transforms raw Reddit discussions and economic indicators into a cleaned, structured, and thematically filtered corpus. The construction process involves three main phases: data collection, data cleaning, and keyword filtering.

4.1.1 Data Collection

As described in Chapter 3, Reddit data were collected across three topical domains: sustainability, consumer behavior, and economic context. This section presents the detailed data volume, subreddit composition, and temporal coverage of the dataset. Data comprising both submissions (initial posts) and comments were collected, covering the period from January 2022 to December 2024. This timeframe was selected to capture recent consumer behavior shifts influenced by economic factors, particularly inflation. Data collection specifically targeted sub-

reddits known for discussions related to the core research themes:

- **Sustainability:** r/sustainability, r/ZeroWaste, r/Green, r/environment
- **Consumer Behavior:** r/BuyItForLife, r/Frugal
- **Economic Context:** r/Economics (with an initial focus on inflation-related discussions, creating an Economics_inflation subset during preliminary stages).

The initial raw data volume collected from these sources was substantial, comprising millions of posts and comments, as indicated by the preliminary counts:

- **Submissions:** Totaling over 100,000 initial posts across the selected subreddits (e.g., 16,215 from BuyItForLife, 13,408 from environment, 57,160 from Economics, 27,660 from Frugal, ...).
- **Comments:** Exceeding 2 million comments (e.g., 995,087 from BuyItForLife, 1,210,256 from Frugal, 1,173,464 from Economics_inflation, ...).

This large initial volume required significant cleaning and filtering to ensure data quality and relevance.

To contextualize online engagement patterns, a set of macroeconomic indicators was retrieved from publicly available sources, covering the same time period as the Reddit dataset (January 2022 to December 2024). These indicators include:

- **Consumer Price Index (CPI-U)** —monthly all-items index from the U.S. Bureau of Labor Statistics.
- **Personal Consumption Expenditures Price Index (PCEPI)** —sourced from the Federal Reserve Economic Data (FRED).
- **University of Michigan Consumer Sentiment Index (ICS)** —a measure of consumer confidence in the economy.
- **Unemployment Rate** —national unemployment levels reported monthly.

The data were standardized to monthly frequency and merged with Reddit discussion data based on the `year__month` key. This enabled comparative temporal analyses, particularly in the Macroeconomic Alignment Module, where shifts in public discourse are compared against real-world inflation signals.

4.1.2 Data Cleaning

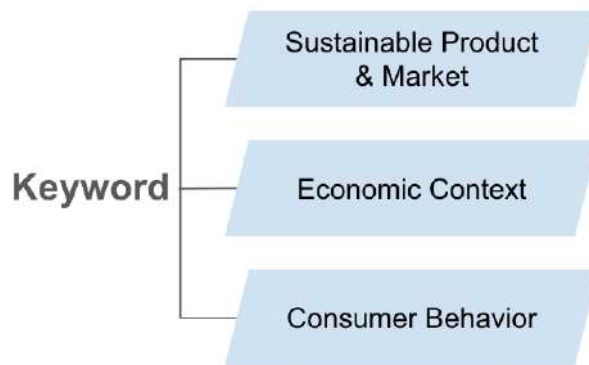
Given the inherent noise and variability in social media data, a multi-step cleaning and pre-processing pipeline was applied to the raw collected data. The objective was to standardize the text and remove irrelevant or low-quality content before thematic filtering. The key cleaning steps were:

1. **Removal of Non-Informative Content:** Submissions consisting solely of URLs were removed. Comments marked as [deleted] or [removed] were excluded. Known bot-generated comments were also identified and removed.
2. **Comment Length Filtering:** Comments with fewer than four words were filtered out, assuming they lacked sufficient substance for analysis.
3. **Text Normalization:** The core text content was normalized by converting it to lower-case and removing special characters, punctuation, and excessive whitespace to create a standardized `clean_text` field.
4. **Handling Missing Values:** Records lacking `clean_text` after normalization were dropped.

After applying these cleaning steps across the targeted subreddits, the dataset size was reduced significantly, retaining more meaningful content. The approximate record counts after cleaning but before keyword filtering were:

- Submissions: about 15,600
- Comments: about 558,000

These cleaned records were then combined into a single dataset, resulting in a total corpus of 573,555 cleaned posts and comments ready for keyword-based thematic filtering.



4.1.3 Keyword Selection

To isolate content relevant to the study’s focus—namely **Sustainability**, **Consumer Behavior**, and **Economic Influence**, a targeted keyword filtering procedure was applied to the cleaned dataset. This step was crucial for minimizing noise from unrelated Reddit content, improving downstream analysis efficiency, and enhancing the thematic focus for sentiment and topic modeling.

The keyword selection process began with the design of a three-tiered thematic structure ?? informed by both literature and research objectives. Each theme was divided into subtopics, such as “Environmental Action” under Sustainability or “Inflation Discussion” under Economic Influence. An initial set of keywords for each theme was manually drafted, then expanded using GPT-4o to suggest colloquial variants, synonyms, and Reddit-specific phrasing, improving coverage beyond formal or academic terminology. For instance, formal terms like “sustainable consumption” were supplemented with phrases such as “trying to be greener” or “reddit recommends.”

To further refine the keyword lists, a set of programmatic tools was applied. Common variants (e.g., “eco-friendly” → “eco friendly”, “ecofriendly”) were automatically generated, and the `pyspellchecker` library was used to correct potential misspellings (e.g., “enviroment” → “environment”). These adjustments ensured a higher match rate for informal or misspelled content frequently found in user-generated posts.

The final keyword sets were used to construct regular expression patterns for each theme,

which were matched against the `clean_text` field. Posts matching one or more themes were retained and assigned corresponding theme tags. The final filtered dataset consisted of **212,841** records, distributed across the three themes as follows (noting that posts could belong to multiple themes):

- **Sustainability:** 52,900 records
- **Consumer Behavior:** 121,055 records
- **Economic Influence:** 38,886 records

The keyword lists used for filtering are summarized below:

Sustainability: *climate, carbon, sustainability, renewable, emissions, energy, waste, plastic, climate crisis, global warming, eco-friendly, low carbon, recycling, reuse, reused, biodegradable, sustainable living, green lifestyle, minimal waste, eco habit*

Consumer Behavior: *vegan, veggie, plant-based, plant based, organic, whole food, meatless, reduce meat, sustainable food, climate-friendly food, buy, bought, purchase, brand, quality, durability, zero waste, long lasting, cheap vs durable, conscious consumer, slow fashion, ethical brand, love, hate, regret buying, worth it, good for environment, bad for planet, feel guilty, eco anxiety, trying to be greener, influenced by, social trend, movement, activism, people like me, peer pressure, reddit recommends*

Economic Influence: *expensive, affordable, worth the price, cheap and good, pricey but lasts, budget friendly, cost vs quality, save money, inflation, cut costs, buy less, spending habits, can't afford, delayed buying, wait for sale, economic pressure*

4.2 Exploratory Data Analysis (EDA)

Before conducting fine-grained sentiment and influence analyses, we first explored the overall patterns in volume and thematic structure within the filtered dataset (`filtered.csv`). This step provided initial context regarding user activity and content focus from January 2022 to December 2024.

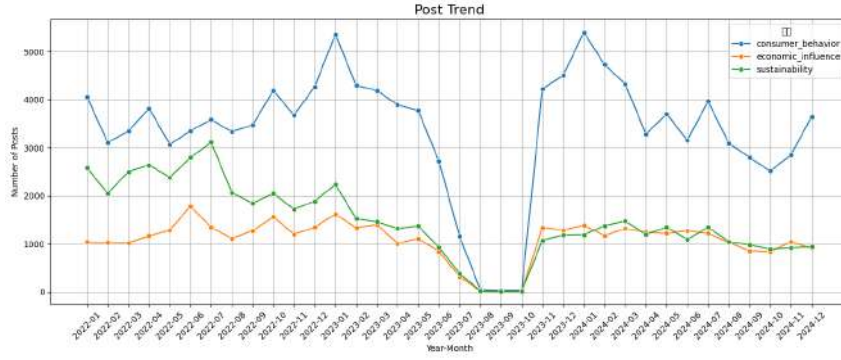


Figure 4.1: Overall Post Trend by Keyword Theme (Jan 2022 - Dec 2024)

4.2.1 Temporal Trend Analysis

To observe how the discussion volume evolved over time, we aggregated the number of posts and comments for each primary keyword theme on a monthly basis. Each post's timestamp was transformed into a `year__month` format, and counts were computed separately for Sustainability, Consumer Behavior, and Economic Influence.

Figure 4.1 shows the resulting time series trends. We observe some fluctuations across themes, with a noticeable dip in July 2023. This corresponds to a known platform-wide protest event on Reddit triggered by changes to the API policy, which significantly reduced user activity. Such events provide important context for interpreting time-sensitive analyses in later chapters.

4.2.2 Semantic Clustering Analysis

To further uncover sub-topics within the broader keyword categories, we applied semantic clustering to each theme separately. This allowed us to identify recurring concepts that may not have been visible through keyword matching alone.

We first encoded up to 5,000 sampled posts per theme using the `all-MiniLM-L6-v2` sentence transformer model, generating dense vector embeddings that captured the semantic meaning of each post. We then used the KMeans algorithm to group the embeddings into coherent clusters. The number of clusters per theme was determined based on manual interpretability. For example, three clusters were selected for the Economic Influence theme.

To label each cluster, we extracted representative sentences close to the cluster centroids and manually reviewed them. In some cases, GPT-4o was used to assist with summarizing and validating the semantic coherence of each group. The final cluster labels are shown in Table 4.1.

Table 4.1: Semantic Cluster Labels Derived from Representative Sentences and LLM Assistance.

Theme	Cluster Labels
Sustainability	Resource & Food Redistribution Climate Action & Impact Creative Reuse Plastic Industry Critique
Consumer Behavior	Economic Hardship Frugal Strategies Quality Durable Goods Thrift Secondhand Anti-Fast Fashion
Economic Influence	Time-for-Value Strategy Frugality Paradox Inflation & Rate Policy

To better understand how these semantic clusters evolved over time, we visualized them using two techniques. First, UMAP was applied to reduce the high-dimensional embeddings into two-dimensional space. This allowed us to generate scatter plots (Figures 4.2, 4.4, 4.6) that reveal the spatial distribution and separation of semantic clusters within each theme. These visualizations help illustrate how distinct the underlying discussion topics are.

Second, to examine the significance of each cluster evolved over time, we aggregated monthly post counts by cluster and plotted the results as time-series graphs (Figures 4.3, 4.5, 4.7). These trend plots provide insight into how certain sub-topics gained or lost attention during the study period, offering a temporal view of shifting user priorities and concerns.

This semantic clustering complements the keyword-level themes by revealing more specific patterns of discourse. These refined sub-topics provide additional granularity for the subsequent sentiment and influence analyses presented in later chapters.

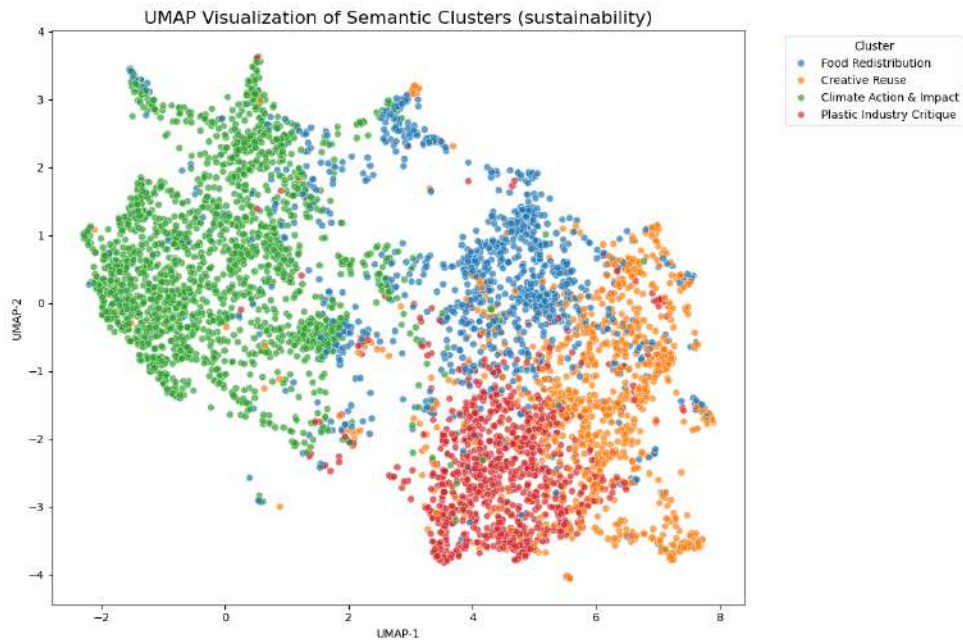


Figure 4.2: UMAP Visualization of Semantic Clusters (Sustainability Theme).

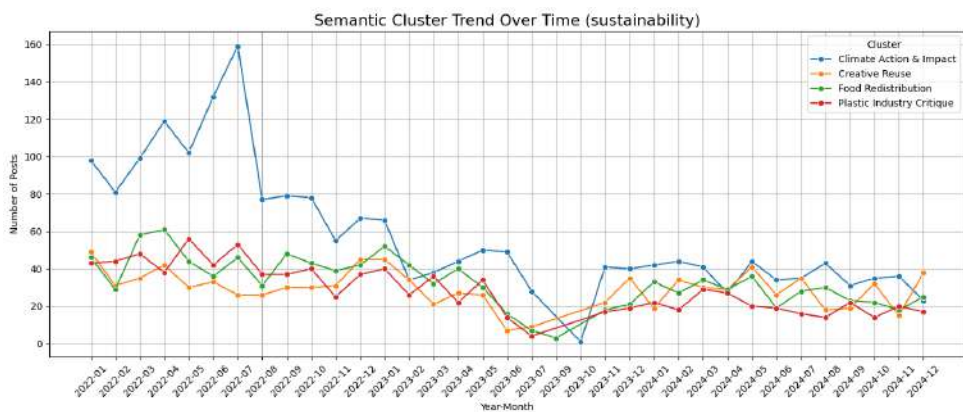


Figure 4.3: Semantic Cluster Trend Over Time (Sustainability Theme).

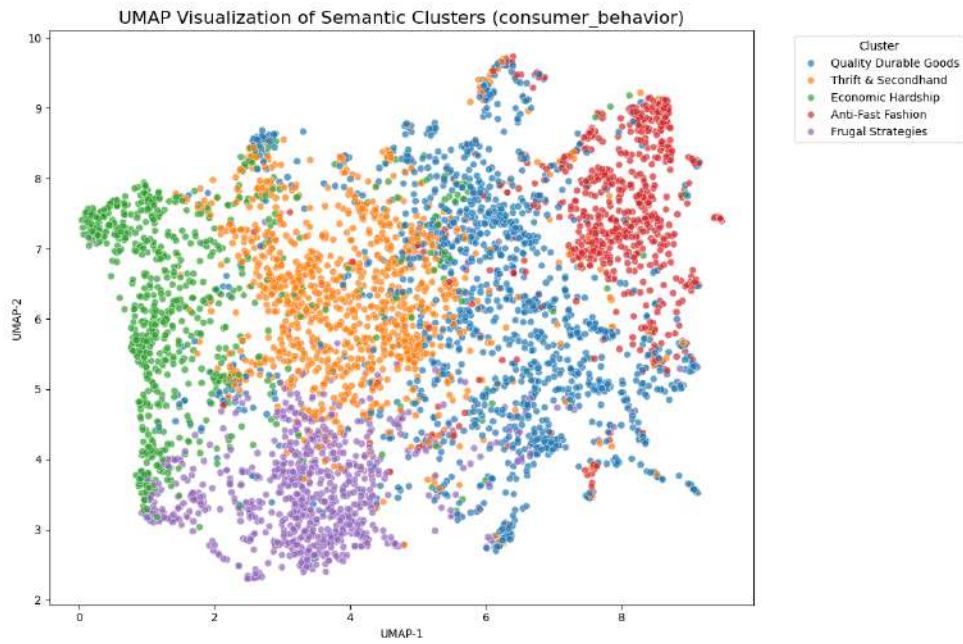


Figure 4.4: UMAP Visualization of Semantic Clusters (Consumer Behavior Theme).

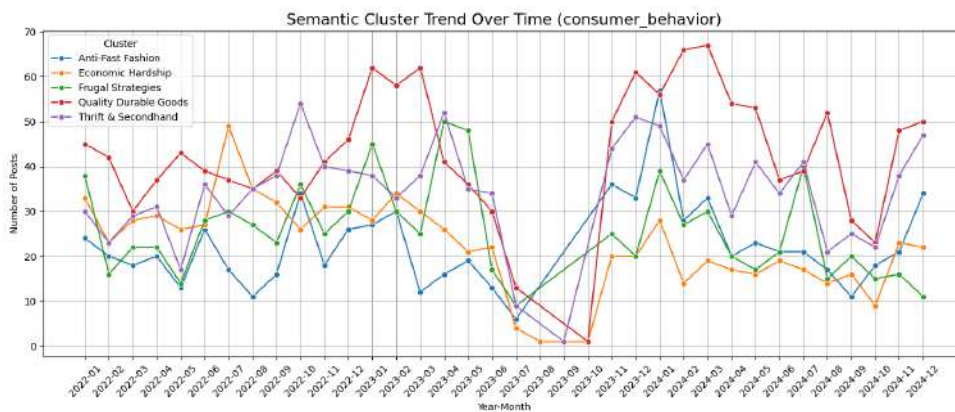


Figure 4.5: Semantic Cluster Trend Over Time (Consumer Behavior Theme).

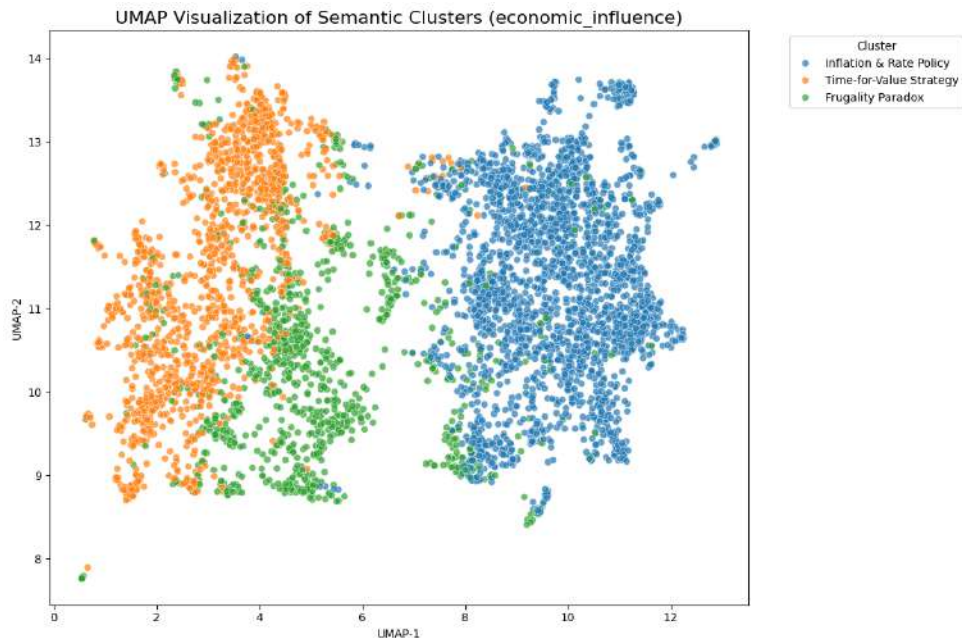


Figure 4.6: UMAP Visualization of Semantic Clusters (Economic Influence Theme).

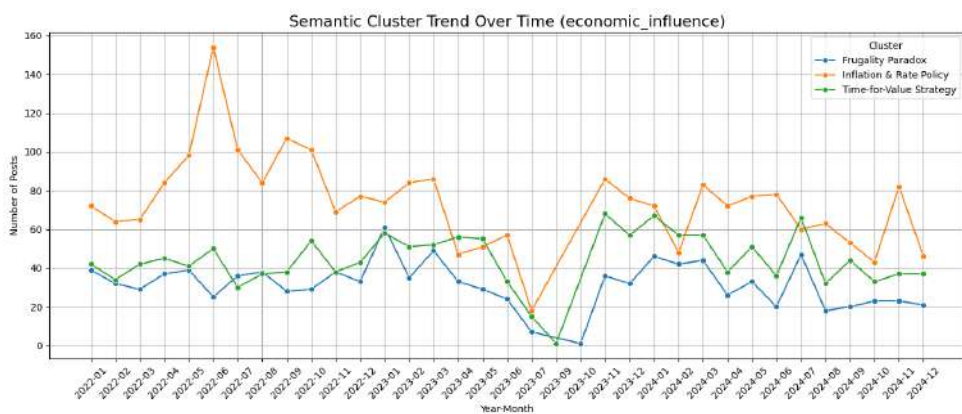


Figure 4.7: Semantic Cluster Trend Over Time (Economic Influence Theme).

4.3 Customer Preference Analysis

4.3.1 Sentiment Analysis

To understand how economic condition shapes public attitudes toward sustainability, we conducted a two-stage sentiment analysis. This approach balances efficiency and depth, allowing us to examine both large-scale emotional trends and the nuanced feelings and intentions behind individual posts.

General Sentiment Classification Using XLNet

Given the size of the dataset (approximately 212,841 Reddit posts and comments), it was not feasible to directly apply computationally intensive LLMs across the entire corpus. Instead, we used `minh21/XLNet-Reddit-Sentiment-Analysis`, a transformer-based model fine-tuned on Reddit data, which is well-suited for large-scale sentiment classification.

The model was run on each cleaned Reddit post using the Hugging Face transformers pipeline with GPU acceleration and a 512-token limit. For each post, the model returned a sentiment label (positive, neutral, or negative) along with a confidence score. Any posts that failed to parse correctly were labeled as "ERROR" and removed from further analysis. We converted the sentiment labels into numerical values: 0 for negative, 1 for positive, and 2 for neutral. These values were stored in a new field called `sentiment_label` and used for subsequent analysis.

To evaluate how each sentiment category was received by the Reddit community, we also computed the average sentiment score for each theme—sentiment pair using the formulation defined in Chapter 3, Section ???. These scores help assess how different types of sentiment align with community response, serving as a proxy for online resonance and validation.

The labeled dataset (with `_sentiment.csv`) allowed us to perform large-scale statistical analysis. We first looked at how sentiment was distributed across the three main themes in the study: sustainability, consumer behavior, and economic influence. We also compared the sentiment distribution between high-engagement posts (top 10% by score) and others. These results

provided a broad overview of online sentiment and helped shape the sampling strategy for the second stage.

Fine-Grained Sentiment and Intent Analysis (LLM)

Sampling Strategy

To gain deeper insights into how people feel and communicate about sustainability during inflation, we applied a second-stage analysis using a more advanced language model. Since running large-scale LLMs across the entire dataset was not practical, we instead analyzed a carefully selected sample of about 6,000 posts using a fine-grained sentiment and intent classification.

To make sure the sample reflected the diversity of the full dataset, we used stratified sampling. The goal was to ensure representation across the three main keyword themes (Sustainability, Consumer Behavior, and Economic Influence), as well as different sentiment categories (positive, neutral, negative) as labeled in the XLNet results. We also included posts from different months to capture how sentiment and intent may have shifted over time. In the end, we collected 6,085 posts, though a few were excluded during API calls due to content restrictions, resulting in a slightly smaller final sample.

To choose a suitable LLM for the second stage of analysis, we compared several models available on Azure AI Foundry (Figure 4.8). The goal was to find a model that offered a good balance between response quality and cost efficiency.

After evaluation, we selected **DeepSeek-V3**, which performed well on benchmark tasks while remaining cost-effective for processing thousands of samples. This made it a practical choice for fine-grained sentiment and intent extraction in this study.

For each post, the model was given a structured prompt and asked to return three types of information in a JSON format:

- a fine-grained sentiment label (such as "frustration," "hope," or "trust"),
- an intent label (such as "complaint," "sharing experience," or "offering advice"), and

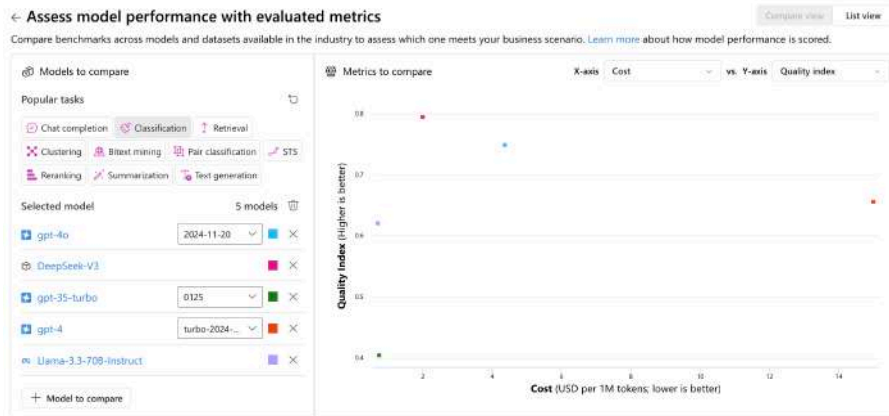


Figure 4.8: Cost versus Quality Index comparison of selected LLMs on Azure AI Foundry, guiding the selection of DeepSeek-V3.

- a general sentiment score on a 3-point scale (negative, neutral, positive) to cross-check with the baseline XLNet results.

This setup allowed us to move beyond general sentiment trends and instead capture more precise emotional patterns and user intentions—such as frustration expressed through complaints, or hope conveyed through advice and encouragement.

Analysis of Fine-Grained Results

The outputs from the LLM analysis enabled a more nuanced exploration of consumer attitudes. The detailed findings from these analyses are presented in Chapter 5. Analyses included:

- **Distribution Analysis:** Examining the frequency distribution of fine-grained sentiments and intents across the different primary themes (Sustainability, Consumer Behavior, Economic Influence) to identify dominant emotional tones and user purposes within each area.
- **Joint Distribution Analysis:** Utilizing heatmaps to visualize the co-occurrence frequency (count-based) and average sentiment score of specific fine-grained sentiment and intent pairs (e.g., how often 'Frustration' pairs with 'Complaint', and whether such posts tend to have high or low scores). This helps identify characteristic patterns of expression.

- **Qualitative Review:** Extracting representative posts corresponding to key or high-engagement sentiment-intent combinations to provide concrete examples and deeper contextual understanding.

Together, these analyses provide a layered understanding of how emotional responses to inflation influence sustainability discourse. By linking sentiment and intent with user engagement, we capture not only what people say, but how their perspectives resonate within the online community.

4.3.2 Topic Analysis

To uncover the main themes discussed across sustainability-related Reddit content, we applied Latent Dirichlet Allocation (LDA), a widely used unsupervised learning technique for topic modeling. This approach helps move beyond surface-level keywords by grouping semantically related words into interpretable topics based on their co-occurrence patterns. Understanding these latent themes is crucial for identifying what consumers are concerned about, especially when economic stress intersects with environmental values.

We used Latent Dirichlet Allocation (LDA) to extract the main discussion topics from the filtered Reddit dataset. This unsupervised learning technique aimed to discover interpretable topics from the textual corpus based on word co-occurrence patterns.

Data Preprocessing for LDA

Before model training, the `clean_text` field from each Reddit post was preprocessed using the Gensim library. The preprocessing pipeline included several steps designed to optimize topic quality: We first tokenized and lemmatized the text, converting each sentence into base-form tokens using the WordNet lemmatizer. Next, we adopted standard stopwords from the NLTK list, along with Reddit-specific filler words such as "im", "thats", or "lol", which often weaken semantic clarity. We then identified frequent multi-word expressions (bigrams) like "climate change" using Gensim's Phrases module, treating them as single tokens to preserve their contextual meaning.

A dictionary was constructed from the tokenized documents, filtering out terms that appeared in fewer than 10 documents or in more than 50% of the corpus. The resulting texts were transformed into a bag-of-words (BoW) representation to prepare for topic modeling.

LDA Model Training and Topic Selection

The LDA model was implemented using Gensim's `LdaModel`. We set the number of topics to $K = 6$, guided by the balance between interpretability and coherence. While topic coherence scores were referenced, the final decision was based primarily on visual inspection using `PyLDAvis`, which allowed us to assess how well-separated and meaningful each topic appeared.

The model was trained with 20 passes over the corpus. We used `random_state=42` for reproducibility and enabled `per_word_topics=True` to facilitate later analyses. Other settings include `chunksize=100`, `update_every=1`, and `alpha="auto"`.

Topic Labeling and Interpretation

After training, we interpreted the topics through a combination of keyword inspection and representative post analysis. For each topic, we reviewed the top-ranked keywords and retrieved the five posts with the highest probability of belonging to that topic. Based on these cues, human-readable labels were manually assigned. To enhance consistency and accuracy, we also used GPT-4o to assist in summarizing the common themes across keywords and post examples.

The six resulting topics and their assigned labels are:

- **Topic 0: Economic Concerns & Companies**
- **Topic 1: Product Experience & Usage**
- **Topic 2: Affordability & Price**
- **Topic 3: Climate Change & Environmental Cost**
- **Topic 4: Sustainable Consumption**

- **Topic 5: Emotional Response & Daily Struggles**

LDA Results and Visualization

The LDA model identified six distinct topics across the Reddit corpus, capturing a range of discussions related to sustainability, affordability, and emotional responses during inflationary times. These topics provide a high-level map of the key narratives that dominate online conversations in this context.

Figure 4.9 shows the relative prevalence of each topic. Among them, discussions related to ”Sustainable Consumption” and ”Economic Concerns & Companies” appeared most frequently, reflecting the dual pressures of environmental values and financial stress that users often navigate.

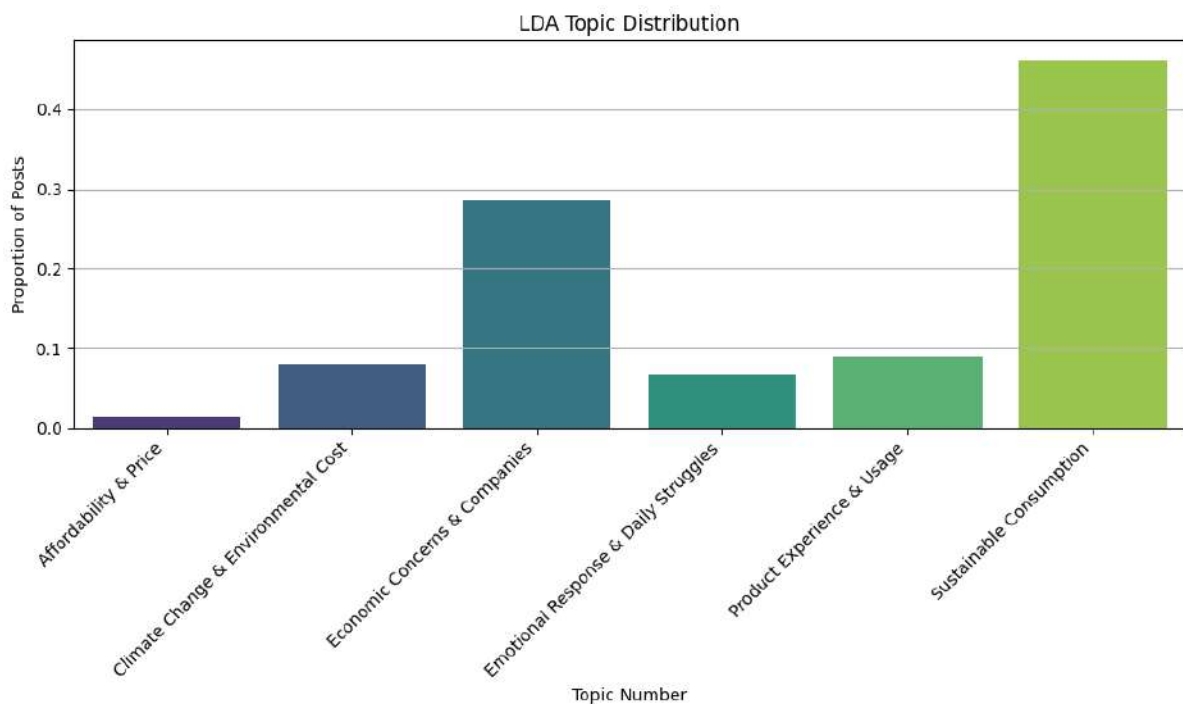


Figure 4.9: LDA Topic Distribution (K=6). Labels reflect the thematic content identified through keyword and representative document analysis.

To further evaluate the distinctiveness and coherence of the topics, we used PyLDAvis to visualize the inter-topic distances and salient keywords (Figure 4.10). This visualization helped ensure that the model captured semantically meaningful and well-separated clusters of discourse,

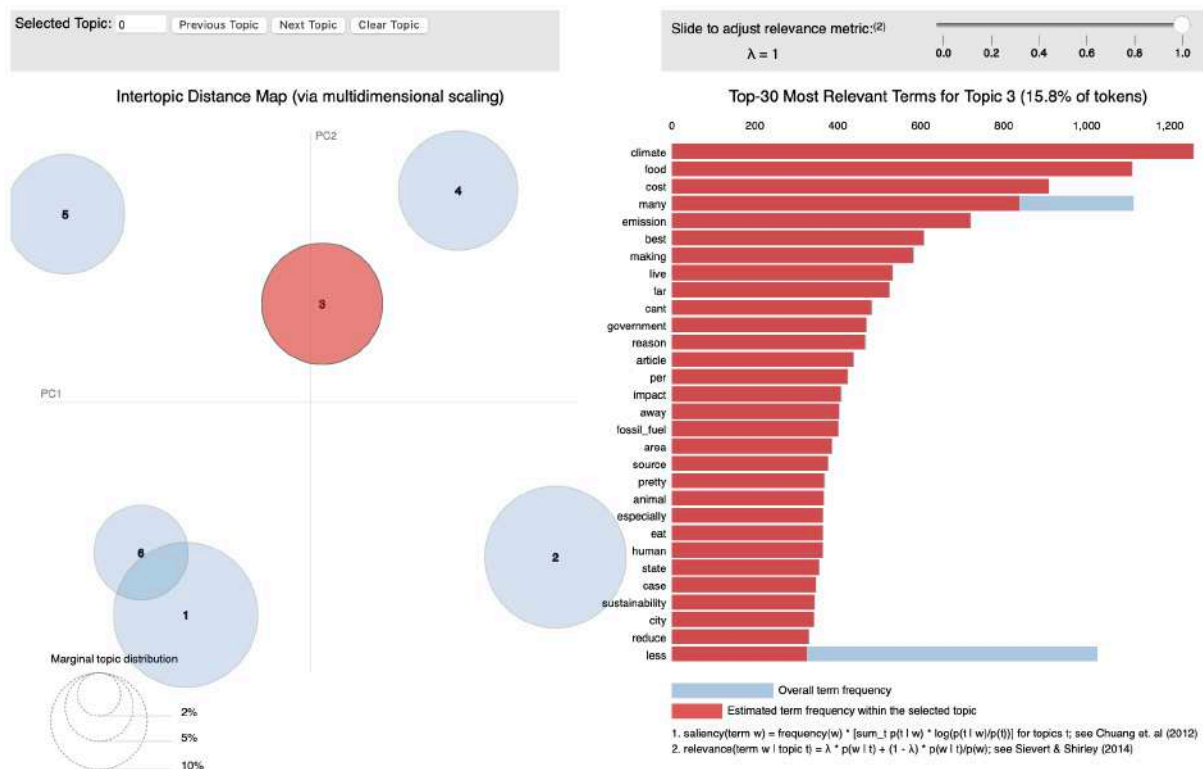


Figure 4.10: Static Representation of PyLDAvis Interactive Visualization for LDA Topics (K=6), showing inter-topic distances (left) and salient terms for a selected topic (right).

rather than overlapping or incoherent themes.

In addition to keyword distributions, we also reviewed representative posts for each topic to interpret their practical meaning and validate the labeling. Table 4.2 presents selected excerpts with high topic probabilities. For example, Topic 0 focuses on structural economic concerns such as corporate pricing behavior and inflation drivers, while Topic 4 emphasizes everyday sustainable practices and long-term frugal habits.

4.4 Social Interaction Analysis

This chapter builds upon the conceptual framework introduced in Chapter 3, translating it into a two-part analytical process. First, we examine **Post Influence**, focusing on which kinds of content tend to generate more community approval and attention. Second, we explore **Author Influence**, identifying users whose contributions consistently receive high engagement, suggesting a role in shaping broader narratives and perceptions.

Table 4.2: Selected Representative Posts for LDA Topics (from representative_posts.txt). Scores are topic probabilities.

Topic Label	Representative Text Snippet
Topic 0: Economic Concerns & Companies	(0.63) it's set by a number of things when supply chain problems happen transportation gets more expensive producer ppi increases... corporate greed also doesn't explain why inflation is up... (0.60) i agree to a limited extent however i disagree with their assessment that increased velocity of money must result in rapid and sustained inflation...
Topic 1: Product Experience & Usage	(0.46) first of all im flad youre ok second...i got run over by a school bus once...i legitimately laughed out loud for a while i didnt expect that sentence to to still end on the frugal note (0.43) its because the dollar supply has gone up by 525x in 2 years what happened trump eliminated the reserve requirement for banks...
Topic 2: Affordability & Price	(0.55) there is a major caveat needed my uncle was a cardiac specialist...some generic cardiac medicines are not safe for some cardiac patients... (0.49) lots of reasons this isnt specifically an environmental reason but since you asked eggs are considered an animal product vegans dont eat animal products...
Topic 3: Climate Change & Environmental Cost	(0.63) game farms can realise some financial compensation for the excess security they need to pay to protect the animals on their lands from poachers in sa... (0.47) compared to agriculture overall animal farming accounts for 18 of the calories contributed but 83 of the land use 58 of greenhouse gas emissions...
Topic 4: Sustainable Consumption	(0.55) i also have trouble hydrating enough without a straw i did buy silicone straws but those are hand wash i have eczema so hand washing destroys my skin... (0.53) new cars have been and will always be a capitallossing proposition this is the way however since i am frugal i also perform all normal maintenance...
Topic 5: Emotional Response & Daily Struggles	(0.55) chea kheang's experience is extreme but by no means unique lot also felt his body changing as he worked on the banana farm... (0.45) it's baffling how more than half of the population is showing signs of depression era thinking when the economy is doing 'well'...

4.4.1 Post Influence Analysis

This section analyzes the characteristics of posts that receive varying levels of community engagement on Reddit, using the post score as a proxy for influence. Rather than predicting exact engagement levels, the goal is to describe how content features correlate with different degrees of post visibility, thereby uncovering patterns associated with effective communication in sustainability discussions.

The influence of each post is quantified using its final score, which reflects the net upvotes (upvotes minus downvotes) it received from the Reddit community. Posts with higher scores are assumed to have greater reach, visibility, or resonance. Due to the heavy right skew in score distribution, where only a small fraction of posts receive very high scores, a tier-based approach is adopted to facilitate meaningful comparisons.

Influence Tier Construction

To compare posts with different levels of influence, the dataset (XLNet Dataset, $N \approx 212,000$) was divided into seven tiers based on percentile thresholds of score: $p_{25} = 14.0$, $p_{50} = 22.0$, $p_{75} = 45.0$, $p_{90} = 111.0$, $p_{95} = 211.0$, and $p_{99} = 781.0$. These labeled groups (from "Very Low" to "Extreme") enable structured comparison across a wide range of influence levels.

Feature Comparison

We compared key post-level features across these influence tiers. These include content type (submission or comment), text length, sentiment (from XLNet), keyword theme, presence of links, and originating subreddit. Statistical tests such as Chi-squared and ANOVA were used to identify significant differences, and visualizations (e.g., bar charts, box plots) were employed to highlight patterns.

Deep Sentiment and Intent Comparison (Sampled Data)

For a subset of posts with fine-grained sentiment and intent annotations from the DeepSeek LLM (Chapter 4.3.1), we further compared the emotional tone and communicative purpose across influence tiers. This provided additional context on how expressive or functional language may relate to post visibility.

Interaction Analysis

We also explored whether the relationship between certain features and post score varies depending on other factors. For example, whether longer posts are more effective as submissions than as comments, or whether negative sentiment leads to more engagement when paired with particular keyword themes. Key feature interactions examined included: $\text{type} \times \text{sentiment_label}$, $\text{text_length_group} \times \text{sentiment_label}$, and $\text{type} \times \text{has_link}$. These were visualized using grouped box plots and tested using ANOVA or Welch’s t-tests where applicable.

Overall, this analysis helps reveal what kinds of content are more likely to resonate with online audiences during discussions of sustainability and economic stress. These findings inform later sections that explore author-level influence and engagement strategies.

4.4.2 Author Influence Analysis

To examine the role of individual users in shaping sustainability discourse, this section implements the Author Influence framework introduced in Chapter 3.4. The analysis quantifies user influence using aggregated engagement scores from their submissions and comments.

The dataset used includes metadata such as author identity, post type (submission or comment), and Reddit score. Records associated with deleted users were excluded. Each author’s influence was computed based on two metrics:

- **Weighted Author Influence Score (AIS):** This metric aggregates the total engagement an author received, computed as a weighted sum of scores from submissions and com-

ments. Following the formula in Chapter 3.4, higher weights were assigned to submissions ($W_{submission} = 0.75$) compared to comments ($W_{comment} = 0.25$), reflecting their generally greater visibility and standalone communicative value.

- **Average AIS:** To account for differences in contribution frequency, the AIS was normalized by the total number of posts and comments, yielding a per-post influence density. This allows high-impact but low-frequency authors to be fairly compared with more prolific users.

Top authors were identified by ranking users within the top 1% according to their Weighted AIS. A similar procedure was applied to the Average AIS, with an additional filter requiring a minimum of five total contributions to ensure robustness.

To facilitate comparative analysis, author-level influence indicators were merged back into the post-level dataset. This enriched dataset allowed comparisons between high-influence authors and the broader user population across several dimensions. Categorical attributes—including keyword theme preferences, sentiment labels, and content type—were evaluated using Chi-squared tests, while numerical attributes, such as average post score, were compared using ANOVA. Visualizations including stacked bar charts and log-scaled box plots were generated to illustrate observed differences.

This analysis highlights which types of users contribute disproportionately to shaping online conversations about sustainability during periods of economic uncertainty. Findings from this section are further interpreted in Chapter 5 to explore how author influence intersects with sentiment, topic, and engagement dynamics.

4.5 Macroeconomic Alignment Module

This section outlines the procedures used to evaluate how Reddit discussions on sustainability and consumer behavior align with macroeconomic trends. The analysis integrates monthly Reddit-derived metrics with U.S. economic indicators to identify potential correlations and time-lagged relationships.

4.5.1 Monthly Aggregation of Reddit Metrics

To enable time-series correlation analysis, post and comment level data were aggregated by calendar month for the period from January 2022 to December 2024. Using the post timestamp, data were resampled monthly to compute the following metrics:

- **Discussion Volume:** Total number of posts and comments per month, along with counts specific to each keyword theme (Sustainability, Consumer Behavior, Economic Influence).
- **Sentiment:** Monthly average sentiment scores across all posts, and theme-specific metrics such as the average sentiment score and positive sentiment ratio within sustainability-related content.
- **Engagement:** Monthly average Reddit score for all content, and separately for sustainability-themed posts.

All resulting time series were standardized to use the first day of each month as the index (e.g., '2022-01-01'), ensuring consistent alignment with economic data.

4.5.2 Economic Indicator Processing

In parallel, monthly U.S. macroeconomic data were compiled and transformed to align with the Reddit metrics:

- **Consumer Price Index (CPI):** Monthly CPI-U values (non-seasonally adjusted) were retrieved from the U.S. Bureau of Labor Statistics. Year-over-year inflation (CPI_YoY_Pct) was computed using the 12-month percentage change.
- **Index of Consumer Sentiment (ICS):** Monthly values from the University of Michigan were used directly as a measure of public economic confidence.
- **Unemployment Rate:** National, unadjusted unemployment rates were obtained from the BLS and used.

- **Personal Consumption Expenditures Price Index (PCEPI):** As a supplementary inflation measure, PCEPI values from the Bureau of Economic Analysis were converted into year-over-year percentage changes (PCE_YoY_Pct).

All economic indicators were indexed to the first day of each month and covered the study period, starting from late 2021 to support lagged calculations.

4.5.3 Data Merging and Correlation Analysis

Reddit metrics and macroeconomic indicators were merged into a unified monthly dataframe using their shared DatetimeIndex. Only months with complete data across all metrics were retained. Minor missing values were handled through forward filling or excluded depending on variable continuity.

From this merged dataset, a **Pearson correlation matrix** was generated to examine linear relationships between Reddit metrics and economic indicators. Corresponding **p-values** were computed to assess the statistical significance of observed correlations.

4.5.4 Visualization and Granger Causality Testing

To aid interpretation, results were visualized using annotated **heatmaps**, highlighting the strength and direction of each correlation. Red (positive) and blue (negative) gradients were used to enhance clarity. For selected variable pairs of theoretical interest, such as inflation vs sentiment monthly line plots were created to compare trends over time.

In addition, **Granger causality tests** were applied to selected time series pairs to explore potential predictive relationships (e.g., whether inflation levels precede shifts in sentiment). Prior to testing, all time series were checked for stationarity using the Augmented Dickey-Fuller (ADF) test, with differencing applied where necessary. While Granger causality does not establish true causation, it helps evaluate temporal dependencies between macroeconomic factors and online discourse patterns.

The results of these analyses are presented in Chapter 5.

4.6 Engagement Driver Identification Module

This section describes how the engagement driver models were implemented and evaluated. Following the conceptual structure in Chapter 3.6, the goal here is to examine how different types of information contribute to predicting post-level engagement.

4.6.1 Feature Sets for Layered Modeling

The experiment follows a layered modeling approach. Each layer adds a specific type of feature, allowing us to compare their individual and combined effects. The definitions below match the six layers introduced in Chapter 3.6:

- **L1_Baseline:** Includes only basic post metadata such as type (submission or comment), text length, and subreddit.
- **L2-1_Customer Preference:** Adds features related to consumer sentiment and topic focus to L1.
- **L2-2_Social Influence:** Adds features derived from the Social Interaction module, including post-level indicators and author-level statistics to L1.
- **L3_Econ:** Adds macroeconomic indicators to L1. These include year-over-year inflation (CPI), consumer sentiment (ICS), unemployment rate, and PCE-based inflation. This layer represents the influence of external economic conditions.
- **L4_Customer Preference + Social Influence:** Combines features from L2-1 and L2-2. This layer captures the joint effect of content preference and community interaction.
- **L5_CS+SI+Econ (Full Model):** Combines all features across previous layers. This is the most complete model and reflects the integrated perspective of content, social, and economic factors.

4.6.2 Model Training and Evaluation

Random Forest was used as the main classifier. It was selected after initial tests showed that it performed better than Logistic Regression and XGBoost on key metrics such as F1-score and AUC, especially when using the full feature set.

Each layer was trained and tested independently. The dataset was split into 80% for training and 20% for testing, with a fixed random seed (42) to ensure consistency across runs. The following metrics were used to evaluate model performance:

- **Accuracy:** Overall proportion of correct predictions.
- **Precision:** Share of predicted high-engagement posts that were truly high.
- **Recall:** Share of true high-engagement posts that were correctly identified.
- **F1-Score:** Harmonic mean of precision and recall.
- **AUC:** Area under the ROC curve, indicating overall classification ability.

4.6.3 Feature Importance Analysis (SHAP)

To understand which features were most influential, we applied SHAP (SHapley Additive exPlanations) to the Random Forest model. SHAP values were computed for the full model (L5), which showed the best performance. These values help identify which features had the largest positive or negative effects on the model's prediction of high engagement. The results were visualized through summary plots and are discussed in Chapter 5.

Chapter 5

Results and Evaluation

5.1 Customer Preference Patterns

This section presents key findings from the Customer Preference Module, focusing on how user sentiment and topic expression shift in response to inflationary pressures. By analyzing Reddit discussions on sustainability, consumer behavior, and economic influence, we observe how emotional tone reveals evolving user preferences under financial constraints.

5.1.1 Sentiment and Engagement Patterns

Using a transformer-based XLNet model, we classified all 212,841 posts into positive, negative, or neutral sentiment categories. While neutral content dominated the dataset, especially in consumer behavior discussions, emotionally expressive posts generally received more engagement.

Figure 5.1 shows that for sustainability-related posts, both positive and negative sentiment led to higher average scores than neutral posts. This suggests that emotionally charged content, whether supportive or critical, generates more attention, aligning with Social Influence Theory (SIT) and the idea that emotional tone shapes peer response.

5.1.2 Fine-Grained Emotion and Intent Analysis

To gain deeper insight, we used a large language model (LLM) to label a stratified sample of 6,085 posts with fine-grained sentiments (e.g., frustration, hope, trust) and user intents (e.g., complaint, suggestion, sharing experience).

Figure 5.2 shows the most frequent combinations of fine-grained sentiment and user intent across the LLM-labeled sample. The most common pairing is clearly *frustration + complaint*,

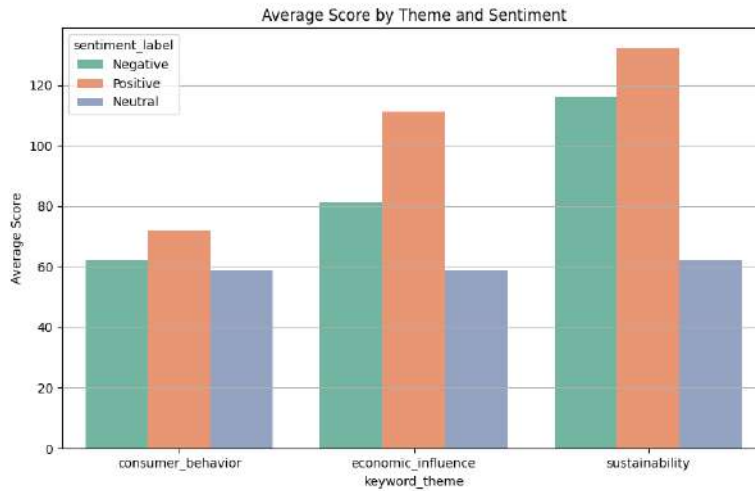


Figure 5.1: Average Reddit Score by Sentiment and Theme (XLNet model).

which appeared nearly 2,000 times in the data. This aligns with findings from Economic Psychology, where emotional stress and blame attribution often emerge under inflationary conditions.

Other frequently observed patterns include *frustration + sharing experience*, *neutrality + sharing experience*, and *joy + praise*. These combinations indicate that, while users often use online platforms to vent dissatisfaction, many also participate by sharing practical experiences or affirming positive encounters. Such expressions may reflect motivations such as social validation and knowledge exchange, as emphasized in Uses and Gratifications (U&G) theory.

5.1.3 Topic Patterns

Topic modeling using Latent Dirichlet Allocation (LDA) identified six dominant themes. Among them, "Product Experience & Usage" and "Affordability & Price" were the most prevalent, indicating that practical considerations and cost evaluation dominate public discourse. Other topics, such as "Climate Change" and "Emotional Struggles," appeared less frequently but carried strong emotional sentiment.

These findings align with the notion that consumer preference under inflation is driven by a blend of cost-consciousness and emotional reflection on broader social issues.

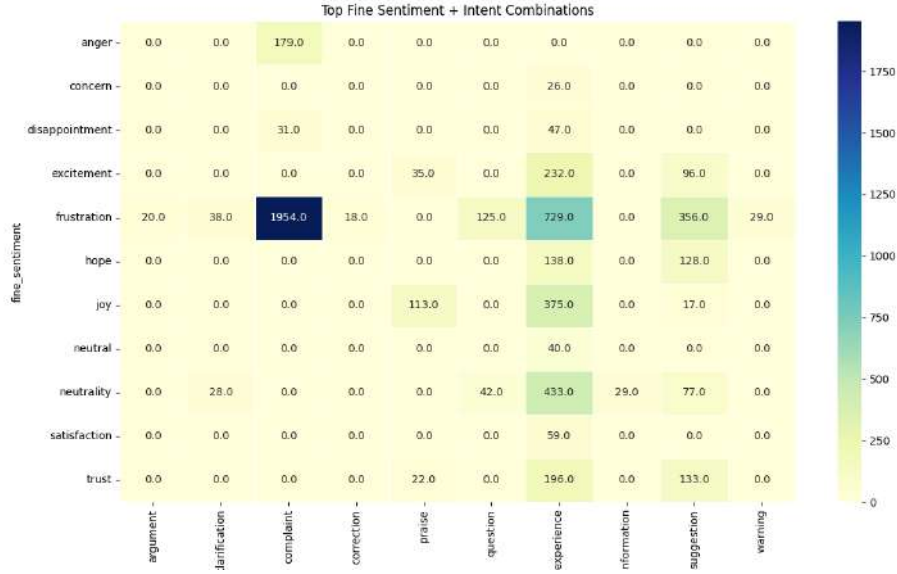


Figure 5.2: Frequency of Fine-Grained Sentiment and User Intent Combinations (LLM analysis, N = 6,085). Darker colors indicate more frequent pairings.

5.1.4 Summary

Overall, user preferences reflect a mix of emotional expression and economic concern. Sentiment patterns suggest that users express both frustration (e.g., inflation-related stress) and constructive engagement (e.g., trust, suggestions), balancing emotional release and social participation. These findings help address RQ1 by highlighting how individual expression adapts to financial pressures, setting the stage for the following analyses of social and economic influence on engagement.

5.2 Social Interaction Analysis

Building on the observed patterns of emotional and topical expression, this section explores how influence operates within Reddit communities. We examine both post-level and author-level dynamics to understand which social cues shape visibility and engagement.

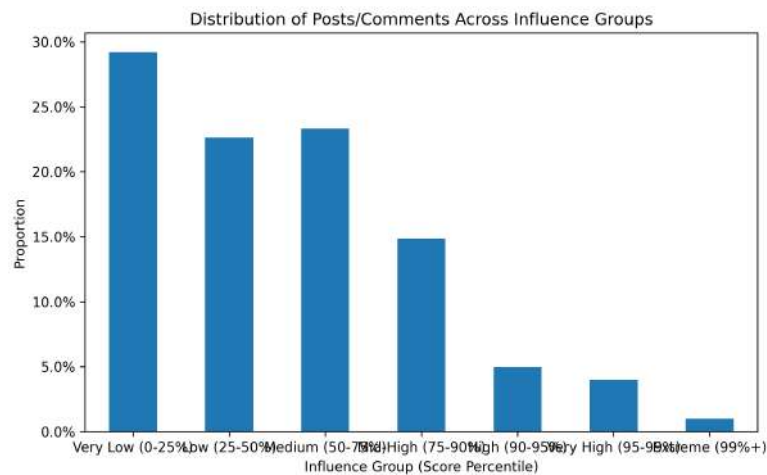


Figure 5.3: Distribution of Posts Across Influence Groups.

5.2.1 Post-Level Influence

Posts were segmented into seven influence tiers based on Reddit score percentiles. As shown in Figure 5.3, engagement is highly skewed, with only 1% of posts in the "Extreme" tier.

Key differences emerged across content features. High-influence posts are more likely to be concise **submissions** containing external links and focused on *sustainability*. These posts often rely less on positive emotion and more on critique or reflection, suggesting that structural clarity and informative value are key engagement drivers under Social Influence Theory.

High-scoring posts commonly combine *experience sharing* or *information sharing* with sentiments such as **trust**, **satisfaction**, or **frustration**. In contrast, low-score content is dominated by *complaints* and *suggestions*. These results suggest that posts which offer practical knowledge or emotionally grounded insights are more likely to gain community approval—again aligning with both normative and informational influence in SIT.

Interactive effects highlight nuanced dynamics. For example, short **submissions** outperform long ones, but text length has less impact on **comments**. Figure 5.6 illustrates that combinations like *trust + suggestion* and *frustration + sharing experience* yield the highest engagement.

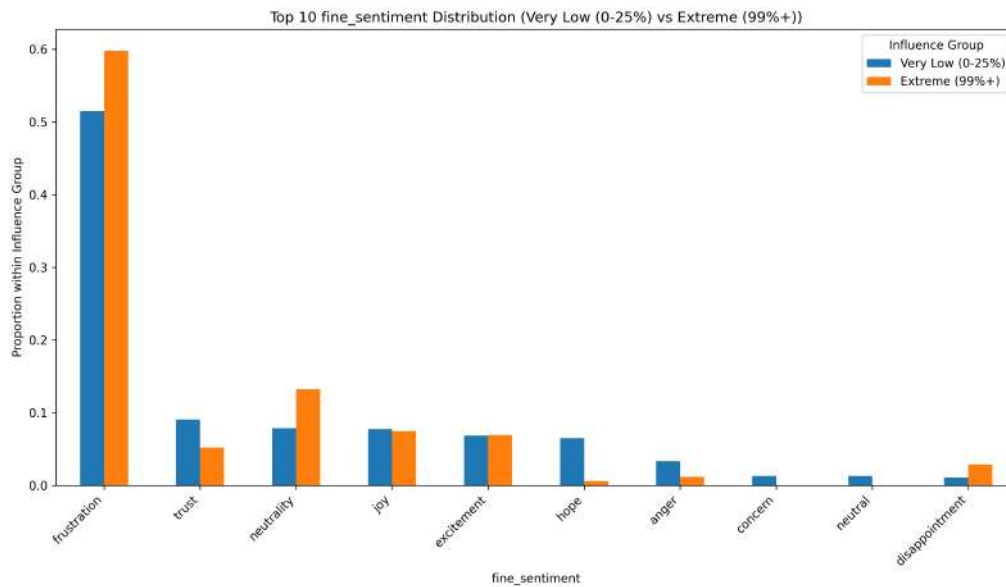


Figure 5.4: Fine Sentiment Distribution in Low vs. High Influence Posts.

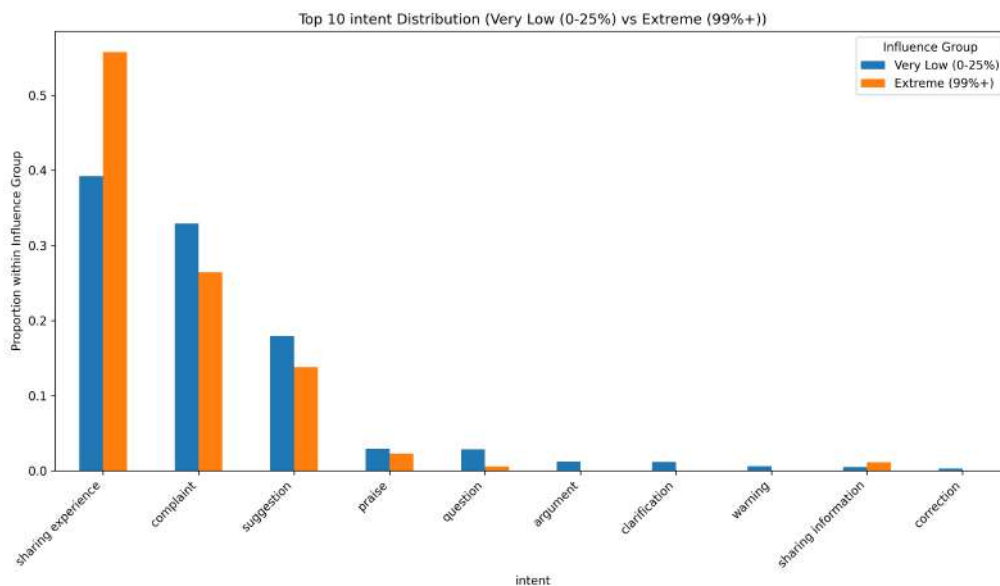


Figure 5.5: User Intent Distribution in Low vs. High Influence Posts.

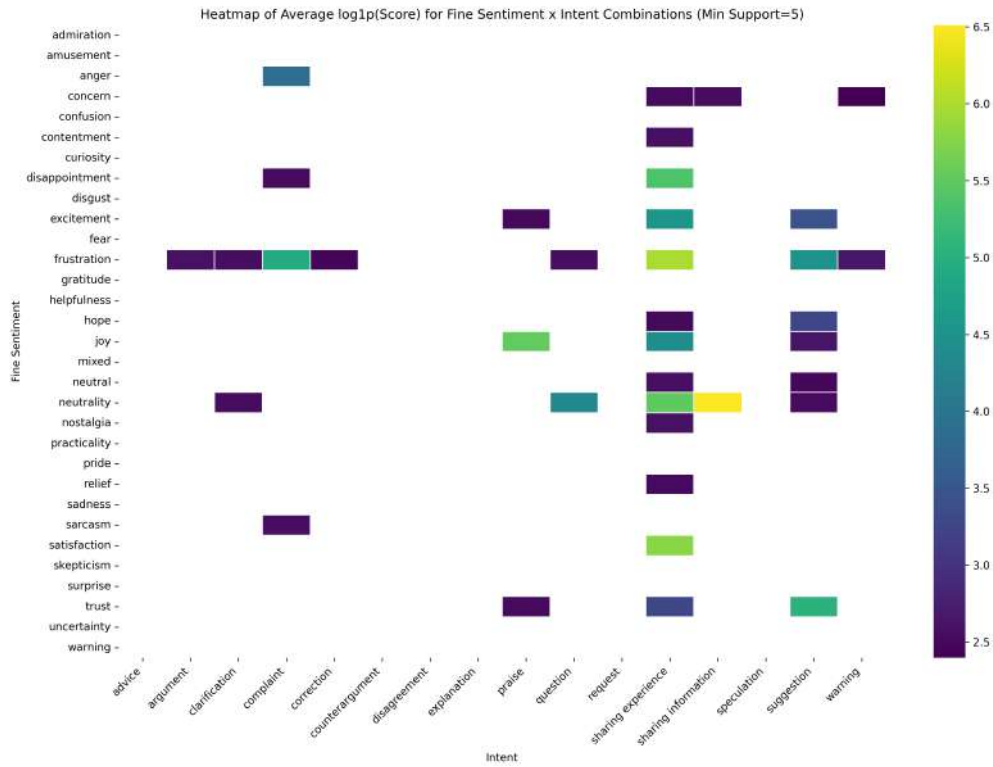


Figure 5.6: Score Heatmap by Fine-Grained Sentiment and Intent (Log Scale).

5.2.2 Author-Level Influence

Influence was also measured at the user level via Weighted AIS, which emphasizes submission impact. Among 88,556 users, the top 1% ($n = 885$) were classified as "Top Authors." As shown in Figure 5.7, top authors contribute a significantly higher proportion of submissions (19.7%) compared to others (2.2%), highlighting their role as discussion initiators. Figure 5.8 shows that they also favor the topic of *sustainability* (36.9%) more than general users (23.6%).

In terms of emotional tone, Figure 5.9 indicates that top authors' content contains more negative (22.2%) and less positive (15.6%) sentiment. This mirrors the pattern seen in high-influence posts.

Finally, as visualized in Figure 5.10, the average score per post for top authors is nearly five times that of other users, with statistically significant difference ($p < 0.001$). These top contributors not only initiate more discussions but also shape thematic focus and emotional framing. Their elevated status positions them as *opinion leaders*, amplifying the reach and

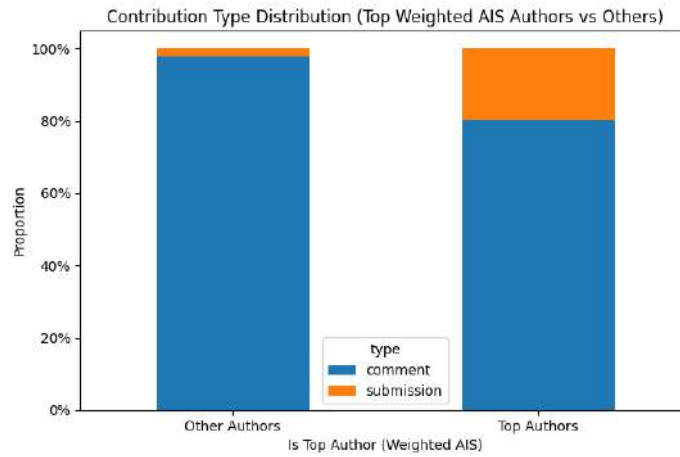


Figure 5.7: Contribution Type Distribution: Top Authors vs. Others.

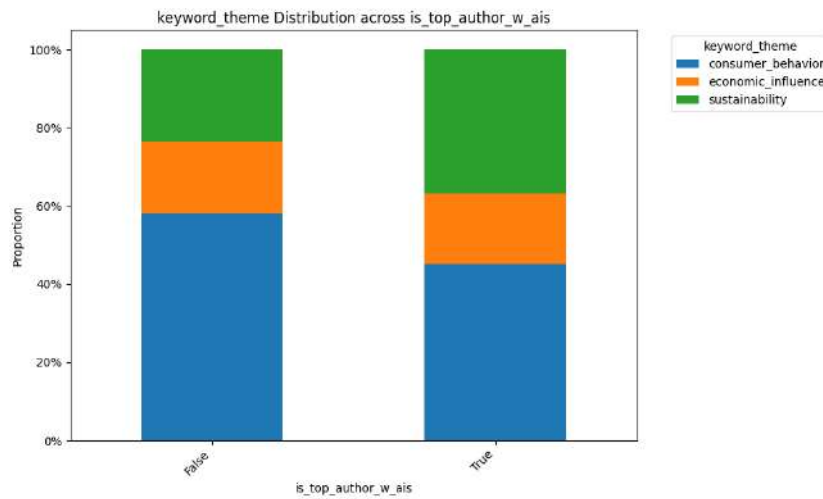


Figure 5.8: Keyword Theme Distribution: Top Authors vs. Others.

framing of sustainability discourse.

5.2.3 Summary

Across both post and author levels, social influence on Reddit is shaped by structural clarity, emotional tone, and contributor identity. High-impact content is not necessarily optimistic, but rather clear, credible, and aligned with informational needs. These findings support **Social Influence Theory**, where both normative reputation and knowledge-sharing shape audience reactions. They also align with **Uses and Gratifications Theory**, which explains why users under stress prefer emotionally resonant and practically useful content.

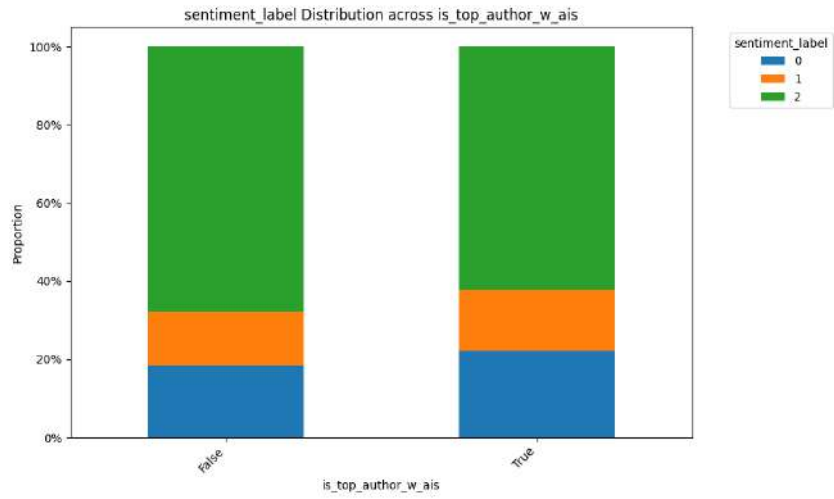


Figure 5.9: Sentiment Label Distribution: Top Authors vs. Others (XLNet).

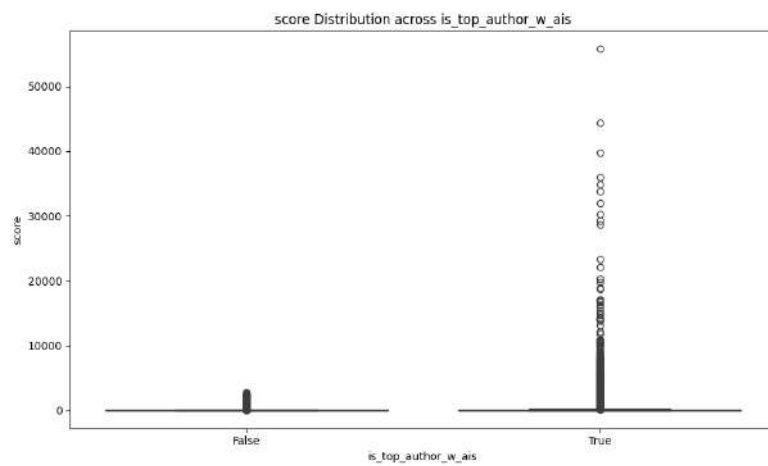


Figure 5.10: Score Distribution of Posts/Comments by Author Tier.

5.3 Impact of Macroeconomic Pressure

This section examines how external economic conditions influence user behavior and sentiment in sustainability-related discussions on Reddit. Drawing from monthly data between 2022 and 2024, we analyze correlations between Reddit-derived metrics and four macroeconomic indicators: Consumer Price Index (CPI), Personal Consumption Expenditures (PCE), Consumer Sentiment Index (ICS), and Unemployment Rate.

5.3.1 Overview of Correlation Patterns

Figure 5.11 shows the Pearson correlation matrix between Reddit metrics (volume, sentiment, engagement) and macroeconomic indicators. While topic volumes are intercorrelated across themes, sentiment and engagement metrics related to sustainability show clearer associations with economic trends.

5.3.2 Key Findings

Inflation (CPI, PCE). CPI shows a moderate negative correlation with average sentiment ($r = -0.37$) and positive sentiment ratio ($r = -0.38$), but a strong positive correlation with average engagement score ($r = 0.70$). This implies that in high-inflation periods, users express more negative emotions in sustainability discussions, yet such posts attract more attention. PCE follows nearly identical patterns, reflecting its close relationship with CPI.

Consumer Sentiment (ICS). ICS is positively correlated with user sentiment ($r = 0.47$), but negatively with engagement ($r = -0.73$). This suggests that when consumer confidence drops, emotionally charged or pragmatically framed content becomes more visible. This aligned with theories from economic psychology.

Unemployment Rate. A moderate positive correlation is observed between unemployment and average engagement ($r = 0.70$), though emotional tone becomes slightly more negative. This further supports the link between economic stress and content visibility.

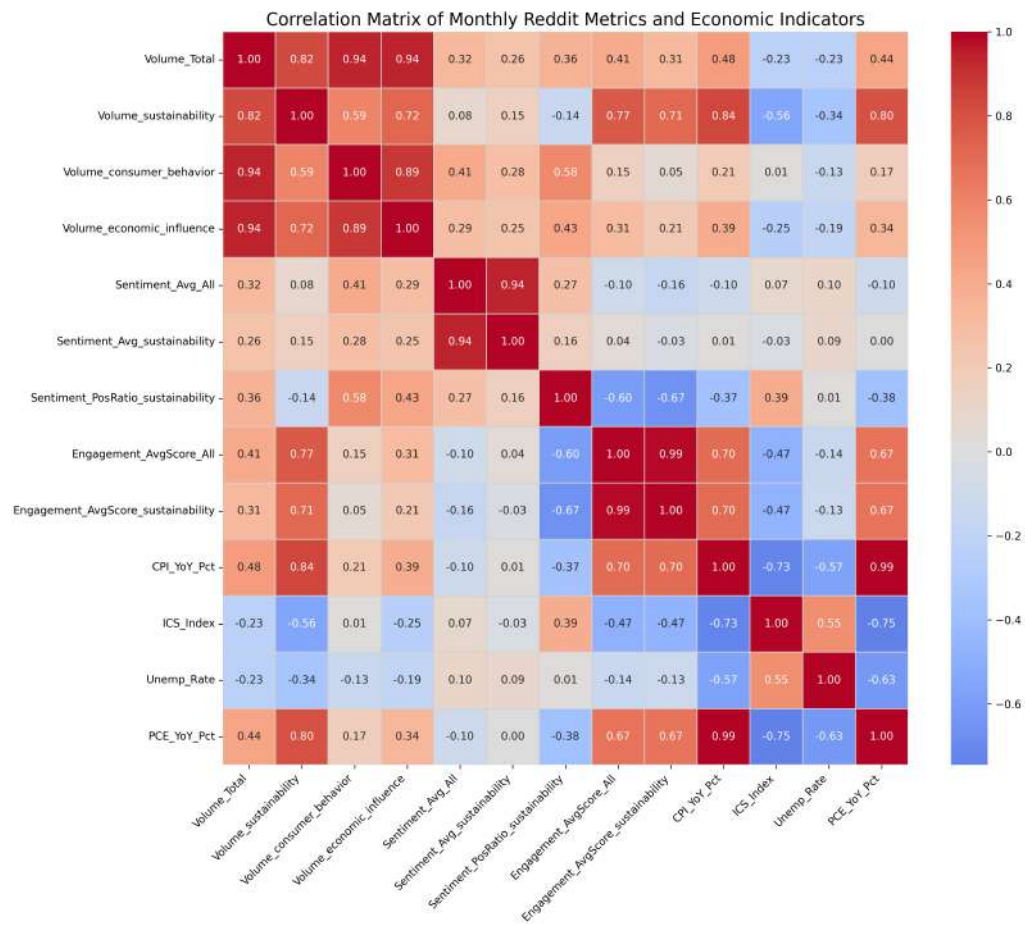


Figure 5.11: Correlation Matrix of Monthly Reddit Metrics and Economic Indicators (2022—2024).

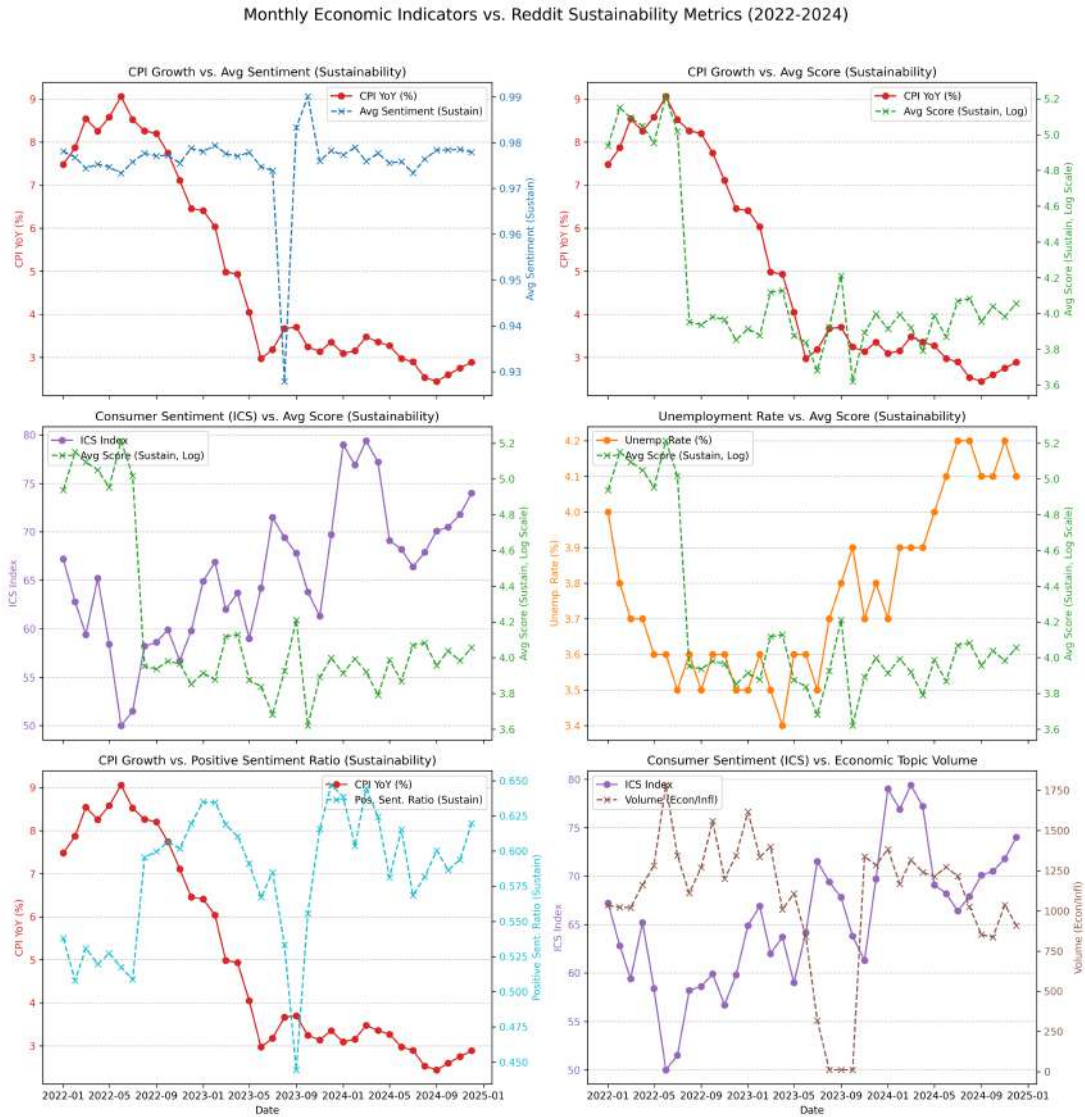


Figure 5.12: Monthly Trends of Economic Indicators vs. Sustainability-Related Reddit Metrics.

5.3.3 Visualizing Temporal Dynamics

Figure 5.12 visualizes trends across selected metrics. As inflation rises (CPI), sentiment declines, while engagement increases. Similarly, as ICS drops, users engage more, possibly reflecting heightened emotional urgency or problem-focused coping. Notably, average engagement score and positive sentiment ratio consistently showed the strongest associations with macroeconomic shifts, particularly inflation and consumer confidence.

5.3.4 Granger Causality Insights

We further conducted Granger causality tests to explore temporal precedence. While sustainability sentiment does not predict inflation, CPI significantly Granger-causes changes in sentiment ($p = 0.022$). This suggests that economic shocks precede shifts in emotional expression, though non-stationarity in CPI cautions against overinterpretation.

5.3.5 Summary

These results reveal a clear link between economic pressure and online behavior. During periods of inflation or low consumer confidence, Reddit users tend to express more negative sentiment—but such posts often receive greater engagement. This pattern reflects concepts from **Economic Psychology**, where stress heightens emotional expression, and aligns with **Bounded Rationality**, as users under constraint prioritize emotionally resonant or pragmatically framed content.

The findings highlight the dual role of inflation as both a constraint and an amplifier of engagement, answering part of **RQ1** by showing that economic signals influence not only sentiment but also what content rises to prominence.

5.4 Final Model Performance and Feature Interpretation

To evaluate how different categories of features influence community engagement, we implemented a layered modeling approach using a Random Forest classifier. This section serves two purposes. First, it evaluates predictive performance across progressively enriched feature sets to test whether the proposed model structure delivers measurable improvement over simpler baselines. Second, it leverages explainable AI to uncover which features contribute most to engagement, providing theoretical insights into how content, social cues, and economic context jointly shape user response.

The baseline model (L1), using only structural metadata, performed poorly. Moderate im-

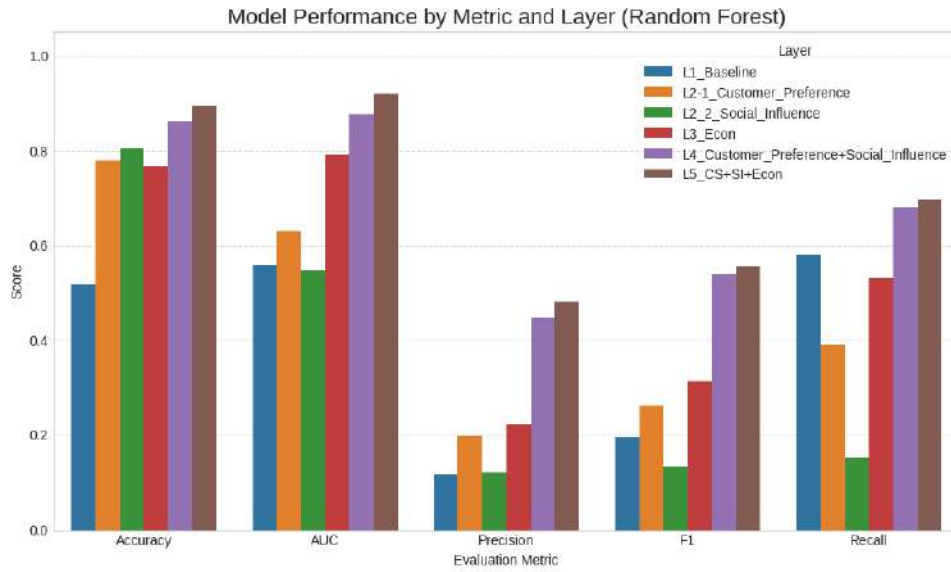


Figure 5.13: Model Performance by Layer and Evaluation Metric (Random Forest).

provements emerged from models incorporating Customer Preference (L2-1), Social Influence (L2-2), or Economic Indicators (L3) independently. However, the most significant performance gain appeared at Layer 4, where combining emotional and social signals, such as sentiment, topic focus, and author reputation, have produced a substantial jump in both F1 and AUC scores. This synergistic effect suggests that socially validated and emotionally resonant content is more predictive of engagement than either signal alone. While Layer 5 (which includes economic indicators) provides a small additional boost, its improvement is incremental, indicating that macroeconomic stress may shape online behavior but primarily through its interaction with social and emotional mechanisms. In practical terms, these findings support the use of Layer 4 features in real-world applications such as content recommendation or message optimization, while also highlighting their theoretical relevance in explaining sustained engagement during economic stress.

To understand what drives these predictions, we used SHAP (SHapley Additive exPlanations) to interpret the final model (L5_CS+SI+Econ). Figure 5.14 highlights the most influential features. Taken together, the SHAP results illuminate how behavioral mechanisms such as trust in credible authors, emotional expressiveness, and cost awareness can manifest through specific feature patterns. These insights not only validate key elements of social influence theory

and bounded rationality, but also extend prior research by showing how such drivers interact dynamically under macroeconomic pressure. In doing so, the model serves not only as a prediction tool but as a theoretical lens to understand engagement behavior in inflation-era sustainability discourse.

The top-ranked predictors came from the social interaction category. In particular, `avg__ais`, `weighted__ais`, and `total__contributions` indicate that author reputation and past engagement are key signals. Posts from active and trusted contributors are more likely to receive attention. These findings align with Social Influence Theory, where audience behavior is shaped by both normative signals (e.g., author credibility) and informational cues. High-AIS users appear to function as opinion leaders, shaping community perceptions and amplifying sustainability narratives.

In terms of content structure, `text__length` had a negative impact, suggesting that shorter posts perform better. This aligns with earlier findings that concise content is more engaging on Reddit. The preference for shorter posts and content from high-AIS authors may reflect bounded rationality. Users simplify decision-making by relying on trusted sources and concise formats, which reduce cognitive effort in evaluating information under uncertainty.

Sentiment-related features, such as `sentiment__score`, also contributed moderately. Though not the most important individually, they help explain influence when considered together with author metrics. From a U&G perspective, users are drawn to content that is concise and emotionally resonant, likely because it fulfills informational and affective needs. The model's emphasis on features like post brevity and sentiment suggests that gratification-seeking behavior plays a central role in shaping engagement.

Finally, macroeconomic indicators—`PCE_YoY_Pct` and `CPI_YoY_Pct`—also ranked in the top 10. While their effects were smaller, they suggest that external economic pressures may affect how users engage with sustainability-related content, especially during inflationary periods. The modest but consistent influence of macroeconomic indicators like CPI and PCE supports insights from economic psychology, suggesting that financial strain can subtly shape user participation by heightening sensitivity to cost-related content or increasing reliance on peer

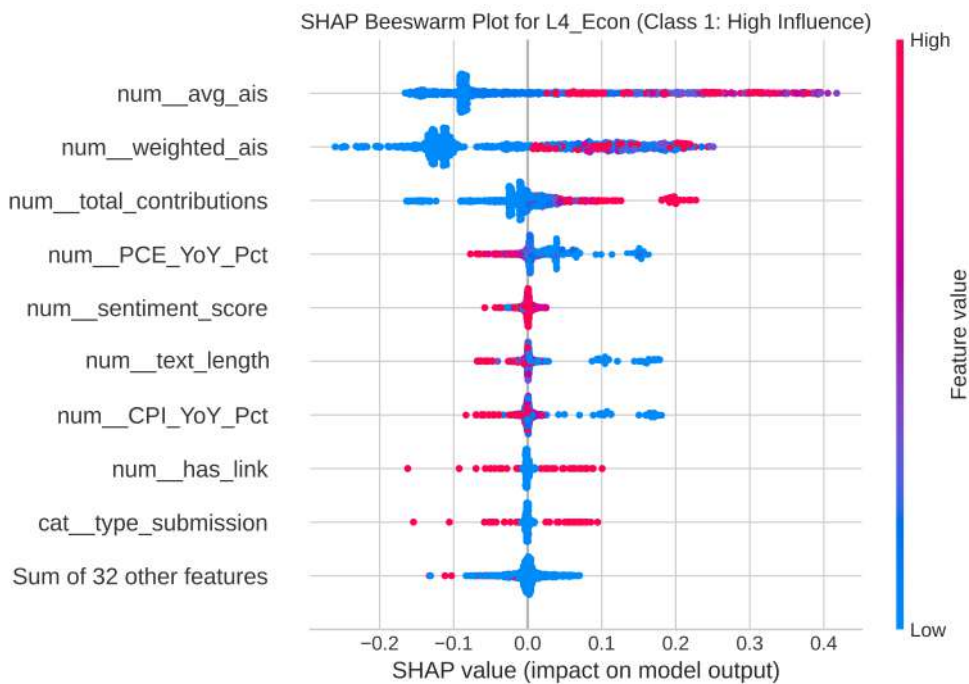


Figure 5.14: SHAP Feature Importance for Final Model (Random Forest).

narratives.

5.4.1 Summary

This chapter shows that the predictive modeling component serves not only to estimate engagement outcomes, but also to clarify the relative importance of different behavioral signals in sustainability discourse during inflation. Rather than pursuing maximum prediction accuracy, the main objective of the model is to provide an interpretable lens through which theoretical constructs—such as emotional tone, social validation, and economic stress—can be evaluated in context.

The use of a layered modeling strategy enables us to systematically examine how different categories of information contribute to engagement. By incrementally adding content features, social indicators, and macroeconomic variables, the model makes it possible to identify the marginal contribution of each factor group. This structured design reflects a key advantage of the system: it does not rely on arbitrary feature selection, but instead aligns the modeling process with theoretically meaningful dimensions.

Explainable AI techniques, particularly SHAP, further enhance this approach by offering transparent insights into which features most strongly influence engagement. The ability to trace model decisions back to specific behavioral variables strengthens confidence in the results and enables closer integration between empirical findings and conceptual reasoning.

The empirical results yield three key findings:

Firstly, **Social signals from author behavior are the most influential.** Author-level metrics such as `weighted__ais` and `avg__ais` consistently rank among the top predictors, suggesting that trust, credibility, and prior engagement history carry significant weight in shaping public attention. These findings are consistent with Social Influence Theory and reaffirm the role of peer dynamics in digital platforms.

Second, **Content structure and sentiment provide additional explanatory power.** Shorter posts and emotionally expressive content (especially those reflecting trust or frustration) are more likely to resonate with users. These patterns align with Uses and Gratifications Theory, indicating that users seek clarity, emotional relevance, and social utility, particularly under conditions of economic stress.

Finally, Macroeconomic indicators play a contextual, supporting role. Although features like CPI and PCE appear among the top contributors in the final model, their influence is more subtle. These variables likely act as background conditions that shape users' sensitivity to certain types of content, rather than directly determining engagement outcomes. This supports theories of bounded rationality, in which users simplify decision-making by focusing on socially validated or emotionally salient cues during periods of financial uncertainty.

Together, these findings offer a multi-layered understanding of online sustainability engagement. By integrating diverse signals through a theory-guided and interpretable model, the study bridges the gap between behavioral research and machine learning practice. The framework not only validates the analytical structure proposed in earlier chapters, but also contributes new empirical evidence about how economic and social forces interact in shaping digital participation. These results address **RQ1** and **RQ2**, offering both theoretical insight and practical implications for understanding influence under inflationary conditions.

Chapter 6

Discussion and Conclusion

6.1 Research Summary

This study explores how consumer engagement with sustainability content is shaped by two competing forces during inflation: economic pressure and social influence. Rather than pursuing predictive accuracy as the end goal, this research treats the model as an analytical lens to identify which content, user, and context level features most influence engagement.

Building on theories from economic psychology and social influence, we constructed a six-module framework to analyze over two years (January 2022 to December 2024) of Reddit discussions. This included sentiment and topic modeling, author influence analysis, economic correlation analysis, and a final predictive model that identifies which content and user-level signals best explain engagement behavior.

Our results show that engagement is not driven by any single factor. Instead, it emerges from a layered combination of post structure (brevity, link inclusion), emotional tone (sentiment and intent), economic context, and especially author reputation. These findings highlight that under financial stress, users continue to seek out emotionally resonant and socially validated content.

6.2 Research Contributions

This study offers contributions from system development, innovation, methodology, and application perspectives:

From a system development perspective, this research constructs a five-module analytical framework that combines content level, user level, and macroeconomic signals. Rather than analyzing engagement as a single-stage prediction, the system integrates multiple theoretical

dimensions into a sequential pipeline, spanning sentiment analysis, social interaction mapping, economic alignment, and interpretable prediction.

In terms of innovation, this study bridges theoretical perspectives from Economic Psychology, Social Influence Theory, Bounded Rationality, and U&G. By empirically examining the interaction between economic headwinds and social tides, the research contributes a novel framing that captures how behavioral mechanisms unfold under inflationary stress. This integrative framing offers a novel way to study sustainability communication not just as a static sentiment, but as a dynamic interaction between individual preferences, community norms, and external constraints.

Methodologically, this study adapts several advanced natural language processing techniques, including transformer-based models (e.g., XLNet) for large-scale sentiment labeling and large language models (e.g., DeepSeek-V3) for nuanced sentiment and intent extraction. The analytical pipeline also incorporates topic modeling (LDA) and semantic clustering to identify key themes, and applies correlation and Granger causality analysis to establish linkages between online discourse and macroeconomic trends. This multi-layered methodology enables scalable, interpretable, and context-aware analysis of user-generated content.

From the application perspective, this study provides practical value for sustainability advocates, policymakers, and communication strategists, this study offers actionable insights. It identifies key characteristics of high-engagement content and shows how these factors can help maintain engagement even in periods of financial stress. The identification of high-impact authors and dominant sentiment-intent combinations enables more targeted communication strategies. Moreover, by revealing how macroeconomic shifts shape user sentiment and topic focus, this study contributes actionable insights into adaptive content design and social media engagement planning. The findings also highlight the role of social media not only as an expression channel but as a behavioral amplifier, capable of reinforcing sustainable values amid competing pressures.

6.3 Research Limitations

While this study provides substantial insights, several limitations must be acknowledged:

Firstly, our analysis focused exclusively on Reddit, a platform with a distinct demographic and interaction model. While Reddit is well suited for long-form, value-laden discourse, the findings may not generalize to platforms driven by visual or short-term content. This platform bias may overrepresent deliberative engagement while undercapturing impulsive or visual-driven reactions common on Instagram or TikTok.

Secondly, the study considered English-language content exclusively, potentially overlooking culturally specific responses and engagement patterns. Future studies incorporating multilingual and multicultural data could enrich understanding of global consumer behaviors.

Thirdly, due to computational and resource constraints, fine-grained sentiment and intent analyses were performed on a limited representative subset rather than the entire dataset. Expanded analysis could reveal additional insights and strengthen the robustness of conclusions.

Finally, multimedia elements such as images and videos, which significantly influence engagement, were not included due to methodological constraints. Future studies should integrate multimedia analysis for a more holistic understanding of online content effectiveness.

6.4 Future Works

Based on the limitations identified, we propose several directions for future research:

Firstly, future research should expand data collection across multiple social media platforms (e.g., Twitter, Instagram, TikTok) to compare cross-platform engagement dynamics and validate the generalizability of findings. Multilingual data collection would further enhance cultural representativeness and facilitate global insights.

Secondly, integrating multimedia analysis, particularly images and videos, could significantly enrich understanding of engagement mechanisms. Employing automated annotation and machine learning techniques for multimedia processing could facilitate this integration.

Finally, exploring newly emerging social platforms, such as Threads, if data access is available, could offer valuable opportunities to analyze evolving consumer behaviors and cross-platform interactions comprehensively. These insights could significantly inform platform specific engagement strategies for sustainability focused communications.

6.5 Conclusion

This study contributes to understanding how economic and social factors interact to shape online engagement with sustainability during inflation. Rather than treating economic stress as a universal dampener, the findings reveal that social validation, emotional tone, and author credibility play stronger roles in sustaining engagement.

In answering RQ1, the results show that social influence partially offsets economic pressure. In addressing RQ2, the model highlights author-level influence and post brevity as the most impactful drivers, underscoring the need for communicators to consider both emotional framing and peer dynamics when designing sustainability messages.

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